

# Cold Crystallization of the Lithium Sublattice and the Warming-Only Resistance Anomaly of Delithiated $\text{Li}_x\text{CoO}_2$

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Chemically exfoliated  $\text{Li}_x\text{CoO}_2$  nanoflakes carry a resistance anomaly that appears only on the warming branch of a thermal loop, leaving the noise near 146 K and passing its half height near 152 K, collapsing through 240 to 245 K in the raw difference between the branches and some ten kelvin higher once that difference is normalized to a cooling baseline that is itself falling, gone in either reading by 260 K, and roughly doubling in a 5 T field while the cooling baseline moves by a few percent. We read the loop as cold crystallization of the lithium sublattice: nucleation and growth are never favorable together along a cooling sweep, so the flake reaches base temperature as a disordered matrix carrying quenched, defect-templated embryos, and those embryos grow as soon as lithium unfreezes, converting the channel to a more resistive ordered phase that a first-order boundary later removes. The unfreezing condition fixes an effective migration barrier of  $0.40 \pm 0.05$  eV, which is the divacancy value in this material, and requires the onset to move by  $12 \pm 1$  K per decade of warming rate at fixed cooling protocol while the collapse stays fixed to better than  $10^{-3}$  K. Density-functional calculations place charge order, a low-spin  $\text{Co}^{4+}$  moment of  $0.97 \mu_B$  and a hard gap on one transition, and leave frustration of that charge order by aperiodic lithium as the surviving account of the resistive contrast. The field-shifted collapse would require the ordered phase to carry the larger magnetization in a unit of order ten correlated moments, but at  $1.4 \pm 1.4$  K/T the shift is consistent with zero at one standard deviation, and our own digitization returns the same sign at  $1.0 \pm 0.5$  K/T in the raw branch difference and at  $0.4 \pm 0.4$  K/T once that difference is normalized, so that chain and the surface electron gas it points to remain conditional. A warming-rate series at fixed cooling protocol decides the central claim with no new instrumentation.

## I. INTRODUCTION

A resistance that depends on which way the thermometer is moving cannot be a property of an equilibrium state. Chemically exfoliated  $\text{Li}_x\text{CoO}_2$  nanoflakes prepared by slow delithiation show exactly that behavior: the cooling trace from room temperature is smooth and featureless, while the warming trace of the same device develops an excess resistance near 146 K, reaches a maximum comparable in size to the cooling baseline itself, and collapses through 240 to 245 K, the raw difference between the two branches passing half its peak height at 244 K and reaching nothing by 260 K [1, 2]. Those collapse temperatures belong to the raw difference; the same excess normalized to the cooling baseline, which is the quantity Fig. 2 plots, falls through its half height about ten kelvin higher, near 254 K, because the baseline it is divided by is itself decreasing with temperature, and every collapse number quoted below names the quantity it belongs to. The loop repeats over thermal cycles without degradation, so no irreversible chemistry is involved. A perpendicular field of 5 T roughly doubles the excess while leaving the cooling baseline within a few percent of its zero-field value, and it pushes the high-temperature collapse upward without moving the low-temperature onset. Those temperatures, the amplitude and the field response are read from the published figures of Refs. [1, 2] by the digitization described with Fig. 2; 146 K is the foot of the rise and not its midpoint, the digitized excess crossing a tenth of its peak height at 149 K and half of it at 152 K, and unlike the collapse the rise gives the same midpoint whichever of the two excesses is used, so that since the onset condition of Sec. IV is a statement about the half-transformed sample it is 152 K and not 146 K that the model is evaluated at throughout, a distinction worth about 0.015 eV in the barrier extracted below. The warming rate

was not recorded, so it is carried throughout as an explicit unknown. Reanalysis of the raw traces would tighten every number quoted below.

Two of those observations already dictate the shape of any explanation. A feature that exists on one branch of a loop and not the other belongs to a transformation that the cooling sweep outruns, which in the language of calorimetry is cold crystallization: the transformation is suppressed on cooling by slow kinetics, is stored in the quenched sample as a metastable state, and completes on reheating as soon as the relevant degree of freedom regains mobility [3–6]. The term belongs to molecular glasses and polymers [3], to metallic glasses [7] and to the chalcogenide phase-change materials whose crystallization on warming is itself read as a resistance step [8–10], and is used here by analogy, no cation or vacancy sublattice of an inorganic layered oxide having been described that way before. A field that acts on the excess and not on the background belongs to a thermodynamic phase boundary rather than to the carriers of the parent metal, because a magnetoresistance that doubled the anomaly by acting on the conduction channel as a whole would have moved the baseline by the same factor; the inference is sound against a bulk-carrier magnetoresistance and not against one localized at the contacts, which Sec. II takes up separately.

The slow degree of freedom is not hard to identify. Lithium in layered  $\text{Li}_x\text{CoO}_2$  migrates at intermediate concentration by a divacancy hop whose barrier first-principles calculations place at 0.41 eV, with a scatter of about  $\pm 0.04$  eV across local lithium and vacancy environments; the isolated-vacancy path is roughly twice as costly, 0.83 eV, and operates only near full lithiation [11–13]. Lithium and vacancy ordering at  $x \simeq 1/2$  is established by diffraction [14–16], while the ordering that cluster expansion predicts at  $x \simeq 1/3$  has never

been observed, its computed boundary lying at or below room temperature [16–19]. Bulk  $\text{Li}_x\text{CoO}_2$  over  $0.46 \leq x \leq 0.71$  shows a first-order anomaly between 155 and 185 K, placed near 155 K and nearly independent of composition in one series of crystals and at 175 to 185 K in another [18, 20], together with a distinct and lower lithium-freezing scale near 130 K below which the magnetization depends on how fast the crystal was cooled and above which  $^7\text{Li}$  motional narrowing and muon-detected diffusion set lithium moving [20–22]; that pair of scales is the bulk counterpart of the two edges analyzed here. In the sister compound  $\text{Na}_x\text{CoO}_2$  the alkali pattern controls the charge order of the  $\text{CoO}_2$  sheets directly [23, 24], so an ionic ordering transition is expected to carry an electronic signature.

This paper builds the minimal model that respects all of those constraints and extracts its parameters. Section II states what the measurement establishes and what the two-probe geometry prevents it from establishing. Section III writes the channel as a two-phase mixture and fixes the size of the transformation. Section IV derives the unfreezing condition, extracts the barrier, and shows why the cooling branch is featureless. Section V treats the field as a thermodynamic variable, converts the observed shift of the collapse into a magnetization difference, and counts how many correlated moments the field must be addressing. Section VI reports the density-functional test of the microscopic ingredients, and Sec. VII lists the measurements that decide the question. An earlier account of this material matched the onset temperature to an equilibrium electronic crossover computed for a nine-site cluster, and we disagree with it for a reason internal to the present model, that an onset set by ionic kinetics moves with sweep rate while no equilibrium electronic scale does.

Throughout, the complete derivations are given in the Supplemental Material [25], whose equations are numbered S1 onward; the main text carries the physical content and the results.

## II. WHAT THE MEASUREMENT CONSTRAINS

The device is a chemically exfoliated single-crystal flake of thickness in the 20 to 60 nm class, contacted by two nickel electrodes across a channel of order  $5 \mu\text{m}$  on  $\text{Si}/\text{SiO}_2$  [1], made and read out by the fabrication and transport pipeline the same laboratory had established on exfoliated oxide nanoflakes [26]. Every contact metal tried on this material is Schottky-like, with a quasilinear current-voltage branch appearing only above about 2.5 V, and the resistance is read from that branch. The measured quantity is therefore

$$R_{\text{meas}}(T; \mathcal{H}) = R_c(T, V; \mathcal{H}) + R_{\text{ch}}(T; \mathcal{H}), \quad (1)$$

where  $R_c$  is the series resistance of the two junctions on that branch,  $R_{\text{ch}}$  is the channel, and  $\mathcal{H}$  denotes the thermal history. The excess

$$\Delta R(T) \equiv R_{\text{meas}}^{\text{warm}}(T) - R_{\text{meas}}^{\text{cool}}(T)$$

cancels every history-independent term in Eq. (1), including any smooth  $R_c(T)$ , but it does not cancel a history-dependent

contact: lithium rearranging under the electrode obeys the same kinetics with the same barrier as the channel mechanism, and would therefore reproduce the branch asymmetry, the amplitude and the rate shift alike. Hysteresis in a layered-crystal device need not be ionic at all, interfacial charge transfer alone having been shown to produce a gate-induced loop in van der Waals heterostructures [27], so neither the presence of a loop nor its size identifies the mechanism on its own. Two observations are consistent with a channel origin without settling it. The anomaly correlates with the delithiation protocol rather than with contact processing, and it survives a change of contact material, a graphene-contacted device from the same batch showing an anomaly above 1 M $\Omega$  with the same phenomenology [1, 2]. Neither observation carries much weight on its own: that device is a single one, its two edges sit nearer 185 and 200 K than the 152 and 240 K quoted here, and a larger anomaly obtained with a different contact metal reads to a skeptic as an effect that scales with contact quality, so the second observation cuts both ways. Only channel-length scaling and a four-probe measurement discriminate, neither has been done, and both are listed in Sec. VII. Joule heating is negligible, since microampere currents at volt-scale bias deposit microwatts into a substrate-mounted flake.

The channel is also a lithium-ion conductor read at 2.5 V across  $5 \mu\text{m}$ , so bias-driven lithium redistribution has to be considered, and slowness is not what excludes it. The Nernst–Einstein relation applied to the same hop model that fixes the barrier below gives a diffusivity of  $2 \times 10^{-10} \text{ cm}^2/\text{s}$  at room temperature, inside the measured chemical diffusivity band of this material, and a drift velocity of  $0.4 \mu\text{m}/\text{s}$  at the naive channel field of 5 kV/cm, so an unscreened field would carry lithium millimeters over a single sweep from 80 to 300 K, a thousand channel lengths. What excludes it is the boundary condition rather than the rate. Nickel electrodes are ion blocking, so lithium moves only until its own space charge cancels the applied field, which it does over a Debye length of 0.04 nm within microseconds above 240 K and at a total displacement of  $3 \times 10^{-14} \text{ m}$ ; the interior of the channel carries no sustained field and no sustained ionic current. Two Schottky contacts in series reinforce the conclusion, since most of the applied 2.5 V drops across depletion regions tens of nanometers wide rather than across the channel. What is not excluded is a Debye-thin lithium space-charge layer at each electrode modulating its barrier, which is the history-dependent contact term of Eq. (1) once more, and a bias-history series, the amplitude measured against read voltage and against dwell time at bias, is the direct test. Every quantitative statement below is a statement about  $\Delta R$  and never about an absolute resistivity, which is the limit of what a two-probe experiment establishes.

The anomaly is also selective in preparation. Flakes delithiated slowly, over a three-day soak in dilute acid, to  $x$  between about 0.37 and 0.48 show it; flakes taken rapidly in stronger acid to nearly the same composition do not, and the onset moves upward as  $x$  decreases [2]. Composition alone therefore does not control the anomaly. What distinguishes the preparations is how much time the lithium sublattice spent near equilibrium, which is the first hint that the physics is kinetic. The drift of the onset with composition is not in conflict

with reading it as a clock, because composition reaches the onset only through the barrier: the divacancy barrier rises as lithium is removed below half filling, and at the sensitivity  $dT_{\text{on}}/dE_a = 397 \text{ K/eV}$  derived in Sec. IV the  $\pm 0.04 \text{ eV}$  environment scatter of that barrier [13] moves the onset by at most about  $\pm 16 \text{ K}$  across the window in play, with the same spread necessarily widening the rise. The batch series of Table S2 in Ref. [25] turns that into a test.

#### A. The lithium sublattice in these flakes

Nothing below requires the parent phase to be structurally featureless, but everything below requires its lithium sublattice to be positionally disordered between 250 and 300 K, and that is what the literature reports at these compositions. The row-ordered monoclinic structure occupies a narrow field around  $x = 1/2$ , bounded by two-phase plateaus near  $x = 0.48$  and  $x = 0.55$  [18]; room-temperature single-crystal diffraction returns the rhombohedral solid solution at  $x = 0.48$  and at  $x = 0.35$ , with the monoclinic distortion explicitly absent at the first of these and only diffuse superlattice intensity too weak to refine [19]; potentiostatically deintercalated single crystals carry no lithium-vacancy superlattice at the electronically stable composition  $x = 0.72$  [28]; and polycrystals at  $x = 0.35$  and  $x = 0.40$  are rhombohedral [18]. The margin is thin even at exact half filling, where the measured order-disorder temperature is only about  $60^\circ \text{C}$ , low for an oxide because charge transfer from lithium to the host screens the lithium-lithium interaction [14, 16]. Four atomic percent of aluminium on the cobalt site removes the hexagonal-to-monoclinic transition altogether [29], and an aluminium-substituted shell reaching only a few hundred angstroms in from the particle surface does the same [30], so chemical and site disorder is established as a destroyer of this order while the separate question of finite size is not, and the exfoliation route used here is known to leave the flakes cobalt deficient [31].

The flakes' own material says the same thing directly. Electron diffraction of as-exfoliated  $\text{Li}_{0.37}\text{CoO}_2$  from this preparation finds only the rhombohedral pattern and its slightly sheared monoclinic companion, the lithium-vacancy superstructure appearing only on annealing near  $200^\circ \text{C}$  [2, 32]; that anneal runs on a clock incommensurable with a  $1 \text{ K/min}$  ramp and is not a measurement of the collapse temperature, but it fixes both the initial state and the direction of the transformation. Independent chemical delithiation by the same acid route to  $x = 0.34$  finds no monoclinic distortion [33], and restacking single layers destroys interlayer registry by construction [34]. What forms on warming is therefore not the clean row-ordered ground state but a partially ordered, domain-textured version of it, which is what the sub- $k_B \ln 2$  latent entropy of Sec. V requires, what the mazed microstructure of Ref. [15] looks like in real space, and what the coexisting ordered and disordered regions imaged in Ref. [35] show.

The sublattice is also mobile in the window where the parent phase has to be disordered.  $^7\text{Li}$  motional narrowing and muon-spin spectroscopy place the onset of lithium hopping between 130 and 150 K [20–22], and the one series that covers

a range of compositions finds it nearly independent of them over  $0.46 \leq x \leq 0.71$  [20], so by 250 K the ions are moving freely and whatever order they hold is equilibrated rather than frozen. The sister compound makes the point quantitatively and from the other side. In  $\text{Na}_{0.8}\text{CoO}_2$  the alkali sublattice orders isothermally below an arrest temperature, with a measured ordering time constant near 40 min at 200 K that lengthens more than tenfold across the 190 to 210 K interval and a quench below 212 K that suppresses the ordering entirely [36]. At a different composition and five orders of magnitude higher in cooling rate, the vacancy liquid that  $\text{Na}_x\text{CoO}_2$  carries above 335 K can be frozen on a nanocalorimeter into what its authors call a sodium vacancy glass [37], so the alkali sublattice of this family arrests over a range of protocols rather than at a single temperature; and the family's high-temperature alkali order-disorder transition, at 427 K in  $\text{Na}_{0.66}\text{CoO}_2$ , is sluggish enough to have been followed at more than one heating and cooling rate [38]. Lithium orders more weakly than sodium for the screening reason given above, which is why sodium cobaltates show well-defined vacancy superlattices where the lithium compound shows none [18, 28].

One alternative slow degree of freedom deserves disposing of explicitly. Delithiated  $\text{Li}_x\text{CoO}_2$  undergoes a stacking transformation from the O3 sequence to H1-3 and then to O1, and that transformation is first order, hysteretic and sluggish, so it would produce the same Kissinger phenomenology as an ionic ordering. It is not available here. The H1-3 phase is measured near  $x = 0.12$  and predicted over 0.12 to 0.19, O1 appears only at  $x = 0$ , and the O3 host is stable throughout  $x > 0.3$  [16, 18, 39, 40], so flakes at  $x = 0.37$  to 0.48 sit a factor of three away from the nearest composition at which a slab glide is thermodynamically available, and the glide barriers are small only once the lithium planes are empty. Were one available it would be unmistakable, collapsing the interslab spacing from 4.81 to 4.52 Å [18] and doubling the  $c$  axis, against the tenth of a percent of metric change that lithium ordering produces; that contrast is the content of the two diffraction entries of Table S2 in Ref. [25].

### III. THE CHANNEL AS A TWO-PHASE MIXTURE

Take the channel to contain two phases: a parent phase P, positionally disordered on the lithium sublattice as Sec. II A documents, with resistivity  $\rho_P$ , and an ordered phase O in which lithium and vacancy order is accompanied by  $\text{Co}^{3+}/\text{Co}^{4+}$  charge order, with resistivity  $\rho_O > \rho_P$ . Let  $\phi$  be the volume fraction transformed. For any microgeometry the mixture conductance lies between the series and parallel bounds on a two-component composite [41, 42], and both bounds give

$$\frac{\Delta R}{R_P} = c \phi (s - 1), \quad s \equiv \frac{\rho_O}{\rho_P}, \quad (2)$$

exactly for the series arrangement with  $c = 1$ , and to first order in  $\phi(1 - 1/s)$  for the parallel arrangement with  $c = 1/s$ ; the derivation and the sense in which  $c \in (0, 1]$  is a geometry factor are given in Eqs. (S1)–(S4) of Ref. [25]. Equation (2)

has the two limits it must have, vanishing when nothing has transformed and vanishing when the two phases conduct alike.

At the zero-field maximum the excess equals the cooling baseline at the same temperature, so  $c\phi(s-1) \simeq 1$ . The transformation is therefore not a perturbation. Either a volume fraction of order unity converts with a modest resistivity contrast, or a smaller fraction converts with a large contrast and is arranged so that the current must cross it. A single reconstructed surface layer on a flake tens of nanometers thick cannot generically produce a relative excess of order unity unless that layer already carries at least half the total conductance, which is precisely what the thickness series of Sec. VII decides. The volume reading is carried below as the simpler of the two and without prejudice, and the two are not interchangeable. Equation (2) itself survives either way, with  $\phi$  read as the transformed fraction of whatever cross-section carries the current, but the per-cobalt normalizations of Sec. V do not: if the transformation is confined to the surfaces, the magnetization and entropy densities in Eq. (9) refer to surface cobalt rather than to all of them, and the thickness dependence of  $\Delta R/R_P$  inverts. Section V keeps both readings open, and Sec. VII gives the pair of measurements that separates them.

Equation (2) takes  $s > 1$  from the measurement and does not explain it, but the same mixture bounds that produced it also bound how large it can be. In the parallel arrangement the excess cannot exceed  $\phi/(1-\phi)$  however large the contrast grows, so an excess of order unity already demands that at least half the current-carrying cross-section transform, and combining that limit with the series one leaves  $s$  between about 2 and 5 with  $\phi$  between 0.25 and 0.75. The contrast the data ask for is therefore modest, which by itself excludes reading the resistivity of the ordered phase as activated across the gap the calculations of Sec. VI return: half of a 0.7 eV gap would give  $s$  between  $10^7$  and  $10^{13}$  across this temperature window, seven to thirteen orders of magnitude from what is measured, and the published traces decay with an apparent activation scale far below any gap in the problem [1]. Both phases must therefore conduct through low-energy weight that a periodic Kohn-Sham gap does not describe, and  $s$  between 2 and 5 says that the ordered phase retains between a fifth and a half of the weight the frozen phase carries. The first-principles calculations of Sec. VI narrow what can be responsible for even that much: the contrast is not a property of which periodic lithium arrangement the ordered phase adopts, since those arrangements are indistinguishable in the one quantity that controls conduction, and the surviving candidate is the frustration that aperiodic lithium occupation imposes on the cobalt charge order of the frozen phase. The kinetic and thermodynamic arguments that follow use only the sign of  $s - 1$  and are unaffected by which microscopic account of its size proves correct.

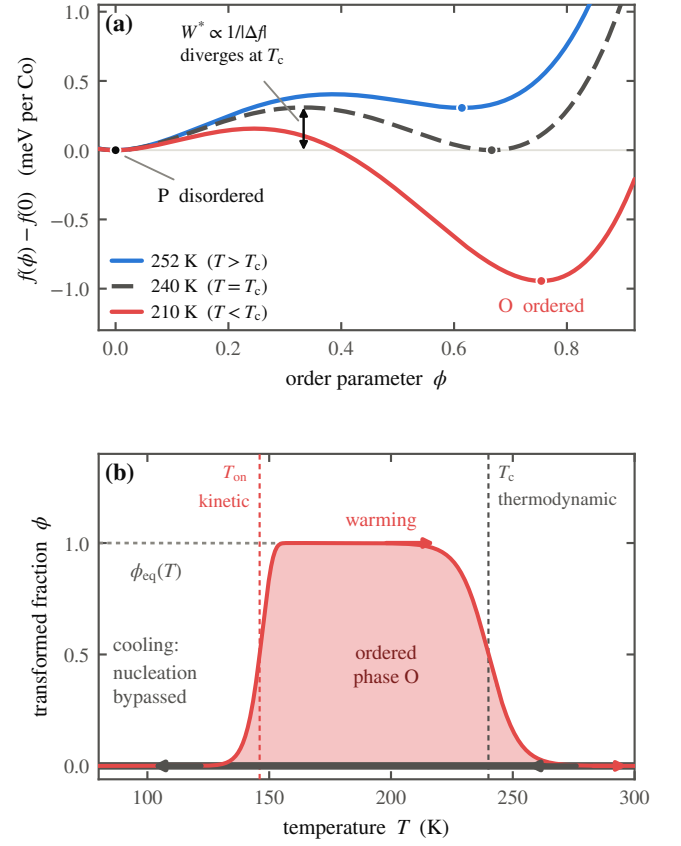


FIG. 1. Schematic of the two-phase free-energy landscape and of the two paths a thermal loop takes through it. The functional is drawn for legibility rather than computed, and no temperature read off it is a result of this work. (a) Free energy per cobalt as a function of the ordering coordinate  $\phi$  at three temperatures spanning the first-order boundary  $T_{\text{ord}} = 240$  K, on an expanded free-energy scale so that the barrier is visible. Above  $T_{\text{ord}}$  the ordered minimum survives as a metastable well up to a spinodal set by the drawn functional, while the disordered parent phase P at  $\phi = 0$  becomes the global minimum; below  $T_{\text{ord}}$  the ordered phase O at large  $\phi$  is the deeper of the two. They are separated by the nucleation barrier  $W^* = \pi\lambda^2/|\Delta f_A|$  of Eq. (S21), which diverges as  $T \rightarrow T_{\text{ord}}$  from below and falls as the undercooling grows. (b) Transformed fraction along a complete thermal loop. The cooling branch stays at  $\phi = 0$  because the region in which a classical nucleus can form and the region in which it can grow do not overlap along the sweep, so the flake reaches base temperature untransformed and carrying the quenched, defect-templated embryos of Sec. IV C. The warming branch rises at the kinetic onset  $T_{\text{on}}$ , where those embryos grow, holds near complete conversion, and falls at the thermodynamic boundary  $T_{\text{ord}}$  where  $\phi_{\text{eq}}$  itself collapses. The onset is a clock reading and moves with sweep rate; the collapse is a phase boundary and does not.



#### IV. KINETICS OF THE TRANSFORMATION

##### A. Unfreezing on warming

Ordering requires lithium to move. Write the minimal relaxation of the transformed fraction toward its equilibrium value,

$$\frac{d\phi}{dt} = k(T) [\phi_{\text{eq}}(T, B) - \phi], \quad k(T) = \nu e^{-E_a/k_B T}, \quad (3)$$

with  $\nu$  an attempt frequency of order a lithium hop-attempt or phonon frequency,  $E_a$  the effective barrier, and  $\phi_{\text{eq}}$  the equilibrium fraction, equal to unity below the first-order boundary  $T_{\text{ord}}(B)$  and to zero above it. Detailed balance is respected because forward and reverse processes share the barrier top, the ratio of their rates being carried entirely by  $\phi_{\text{eq}}$ . Section IV C explains why Eq. (3) with a single effective barrier is legitimate on the warming branch and not on the cooling branch.

Start the warming sweep in the quenched state,  $\phi = 0$  with  $\phi_{\text{eq}} = 1$ , and let  $T(t) = T_0 + rt$ . Integrating Eq. (3) and evaluating the resulting integral by Laplace's method at its upper endpoint, which is controlled by the large parameter  $E_a/k_B T \approx 31$ , gives

$$\phi(T) = 1 - e^{-\Theta(T)}, \quad \Theta(T) \simeq \frac{\nu}{r} \frac{k_B T^2}{E_a} e^{-E_a/k_B T}, \quad (4)$$

with the subleading correction of order  $2k_B T/E_a$ , a few percent here; the steps are Eqs. (S6)–(S11) of Ref. [25]. The dimensionless  $\Theta$  is the number of successful rearrangement attempts accumulated along the sweep, and it scales inversely with the warming rate, as it must: sweeping twice as fast halves the time spent at every temperature.

Defining the onset  $T_{\text{on}}$  by the rise midpoint,  $\Theta(T_{\text{on}}) = \ln 2$ , and solving for the barrier gives

$$E_a = k_B T_{\text{on}} \ln \left[ \frac{\nu k_B T_{\text{on}}^2}{\ln 2 E_a r} \right], \quad (5)$$

an implicit relation solved by iteration in two or three steps. With  $T_{\text{on}} = 152$  K, which is where the digitized excess reaches half its peak height and therefore where the condition  $\Theta = \ln 2$  applies, an attempt frequency  $\nu = 10^{13} \text{ s}^{-1}$ , and a warming rate  $r = 1 \text{ K/min}$ , which is a typical cryostat value carried here as an explicit unknown because the rate was not recorded, Eq. (5) returns 0.469 eV if the growth velocity is taken to have saturated at the onset. It has not saturated there. At 152 K the thermodynamic factor  $1 - e^{\Delta f/k_B T}$  of Eq. (7) still stands at 0.12, and the transformation on the warming branch is growth from a fixed areal density  $n_0$  of seeds rather than the single-exponential relaxation of Eq. (3), so the condition that fixes the barrier is  $\pi n_0 R_w^2(T_{\text{on}}) = \ln 2$ , with  $R_w$  the radius a growth front sweeps along the ramp. Retaining both corrections lowers the barrier to 0.403 eV at  $n_0 = 10^{-2} \text{ nm}^{-2}$ , the two quantities entering together as  $E_a = 0.403 + 0.015 \log_{10}(n_0/10^{-2} \text{ nm}^{-2}) \text{ eV}$ , so the barrier has to be quoted as one member of a pair and never on its own. Propagating the uncertainties gives

$$E_a = 0.40 \pm 0.03 (\nu) \mp 0.03 (r) \pm 0.02 (T_{\text{on}}) \pm 0.01 (n_0) \text{ eV},$$

per decade of attempt frequency, per decade of warming rate, for  $\pm 7$  K on the onset and per decade of seed density, which combine in quadrature to  $E_a = 0.40 \pm 0.05 \text{ eV}$  with a corner-to-corner range of 0.28 to 0.48 eV; the accompanying  $n_0$  runs from  $10^{-4}$  to  $10^{-1} \text{ nm}^{-2}$ , which is seeds 3 to 100 nm apart, or  $10^2$  to  $10^5$  of them per square micrometer.

That value lands on a mechanism rather than merely inside a band. The divacancy hop that carries lithium at intermediate concentration has a computed barrier of 0.41 eV with an environment-dependent scatter of  $\pm 0.04 \text{ eV}$ , and an average over the composition window  $x = 0.37$  to 0.50 that these flakes occupy which has been read as low as 0.36 to 0.39 eV, against 0.83 eV for the isolated-vacancy path that operates only near full lithiation [13]; the extracted 0.403 eV sits between those two readings rather than on either of them, which is as much as should be claimed for it, since what the onset locates is the divacancy family and not one digit inside it, while the isolated-vacancy path is excluded by a factor of two on any reading. The identification should not be pressed hard. With  $\nu$ ,  $r$  and  $n_0$  all free, the quoted interval very nearly fills the range of plausible lithium barriers in this material, so the coincidence has little discriminating power until the warming-rate series of Sec. VII pins the slope. What it does support is reading the onset as an ionic clock rather than as an electronic energy scale.

##### B. The rate shift, and why it is parameter-free

Because  $E_a$  enters Eq. (5) only inside a logarithm, the onset is insensitive to the prefactors and sensitive to the sweep rate in a way that is easy to measure. Differentiating the condition  $\Theta(T_{\text{on}}) = \ln 2$  at fixed  $E_a$  and  $\nu$  gives

$$\frac{dT_{\text{on}}}{d \ln r} = \frac{k_B T_{\text{on}}^2}{E_a + 2k_B T_{\text{on}}}, \quad (6)$$

which is the Kissinger relation for a first-order rate process [43]. At  $T_{\text{on}} = 152$  K and  $E_a = 0.40 \text{ eV}$  this is 4.7 K per  $e$ -fold. The thermodynamic factor of Eq. (7) carries its own temperature dependence at the onset,  $d \ln f / dT = 0.017 \text{ K}^{-1}$  there, which steepens the slope by 8.3 percent, slightly larger than the  $2k_B T/E_a$  correction the Laplace evaluation already retains and of the opposite sign; keeping both gives 11.6 K per decade of warming rate, and across the barrier range admitted above the prediction spans 10.8 to 12.6 K per decade, which is  $12 \pm 1 \text{ K}$  per decade to the precision the model supports.

That prediction holds only if the cooling protocol is held fixed while the warming rate is varied. Should the cooldown be slaved to the warm-up, athermal seeding leaves the slope where it is, but thermally born seeds would thin as  $n_0 \propto 1/|r|$  and steepen the shift to about 17.4 K per decade, so a series run without that precaution measures a number whose interpretation depends on the very seeding mechanism the series is meant to probe. Subject to the protocol, the shift is parameter-free in the sense that matters: the same differentiation applied to a transformation whose exponent is  $\Theta^n$  rather than  $\Theta$  returns Eq. (6) unchanged, because the factor  $n$  multiplies the rate dependence and the temperature dependence alike and cancels,

which the algebra of Eq. (S17) of Ref. [25] shows and which we have confirmed numerically for  $n = 1$  and  $n = 2$ . The predicted shift therefore does not depend on whether the transformation is growth-limited from quenched seeds or governed by continuous nucleation, which makes it the sharpest single test of the whole picture.

The width of the rise is not equally insensitive. Growth in two dimensions from a fixed density of seeds is quadratic in  $\Theta$ , and at this barrier such a transformation spans its 10 to 90 percent range over 7.7 K, from 147.4 to 155.1 K. The measured width,  $19^{+6}_{-9}$  K, sits well above that, and a spread of barriers of about  $\pm 0.044$  eV closes the gap at the sensitivity  $dT_{\text{on}}/dE_a = 397$  K/eV, which compositional inhomogeneity of the kind seen across batches [2] would naturally supply. The same spread is what limits how far the onset can move with composition, so the width of the rise and the batch dependence of the onset are one statement rather than two. Strongly cooperative kinetics, which would sharpen the rise, is not required by the data and is mildly disfavored by it.

### C. Why the cooling branch is featureless

Equation (3) is symmetric under reversal of the sweep and cannot by itself produce an anomaly on one branch only, which is the step most discussions of thermal hysteresis leave out. Applied to the cooling branch, Eq. (3) would predict that the sample transforms immediately below  $T_{\text{ord}}$ , where the relaxation time  $\nu^{-1} e^{E_a/k_B T}$  is at most  $3 \times 10^{-4}$  s across the admitted barrier range and shorter than  $10^{-4}$  s at its center. The asymmetry therefore cannot come from a single rate; it comes from the two-step structure of a real first-order transformation, in which a nucleus must appear before anything can grow.

For two-dimensional ordering in the lithium plane the nucleation rate per unit area and the growth velocity are

$$\begin{aligned} I_n &= I_0 e^{-[E_a + W^*(T)]/k_B T}, \\ v &= v_0 e^{-E_a/k_B T} (1 - e^{\Delta f/k_B T}), \end{aligned} \quad (7)$$

with  $\Delta f = f_0 - f_p$  the free-energy difference per cobalt, negative below  $T_{\text{ord}}$ , and  $W^*(T) = \pi \lambda^2 / |\Delta f_A(T)|$  the barrier to forming a critical disk of ordered phase,  $\lambda$  being the domain-wall line tension and  $\Delta f_A$  the free-energy difference per unit area. The growth velocity vanishes at  $T_{\text{ord}}$ , as it must, and saturates at  $v_0 e^{-E_a/k_B T}$  deep in the ordered regime. Feeding Eq. (7) into the Kolmogorov–Johnson–Mehl–Avrami construction for the transformed fraction [44–47], given as Eqs. (S23)–(S25) of Ref. [25], yields the asymmetry directly.

The two factors in Eq. (7) pull in opposite directions, and that opposition is the whole argument, but the place where they cross has to be located correctly. Nucleation needs a large undercooling, because  $W^*$  diverges at  $T_{\text{ord}}$  and falls only as the driving force grows; growth needs a high temperature, because the mobility factor  $e^{-E_a/k_B T}$  collapses on cooling. The nucleation rate, however, carries that same mobility factor, so it is not large at the bottom of the sweep: below about half the collapse temperature  $I_n$  increases monotonically with temperature, and nucleation at base temperature is the least

likely event in the problem rather than the most. The only window on the cooling branch in which a classical nucleus can form at all is the nose of the transformation curve, which for every admissible line tension sits between 178 and 211 K. A nucleus born there sweeps between 0.6 nm and 40  $\mu\text{m}$  through the remainder of a 1 K/min cooldown, while the warming branch needs its seeds to grow only 1.7 to 20 nm to reach half conversion at 152 K, so a nucleus born at the nose would transform the flake on the way down and destroy the very asymmetry it was invoked to explain. The requirement is quantitative and nearly independent of the barrier: every seed present at base temperature must have been born below  $T_{\text{on}} - 9$  K, which is below about 143 K.

Classical nucleation cannot put them there. Forcing the maximum of  $I_n$  below 143 K demands  $W^*/k_B T \approx 60$  at that temperature, which suppresses the rate by  $e^{-60}$ , and a scan over the barrier from 0.34 to 0.45 eV, the attempt frequency over three decades, the latent entropy from 0.1 to 0.69  $k_B$ , the line tension over four decades and the cooling rate from 1 to 60 K/min leaves the delivered seed density short of what the warming branch requires by five to eight decades at 1 K/min, and by 1.9 decades in the single most favorable corner. Heterogeneous nucleation does not repair it, since a potency factor multiplying  $W^*$  is the same one-parameter family as a reduced line tension while its prefactor is smaller by the areal density of the sites themselves, so the shortfall grows by about a decade rather than closing. Formal closure would need a cooldown of order  $10^3$  K/min, which is a plunge quench and not a cryostat.

What the model needs, and what we adopt as an assumption rather than derive, is that the seeds are not thermal at all. Any barrier of the classical form  $W^* \propto 1/(T_{\text{ord}} - T)$  is excluded by the argument above, so the escape has to break the functional form: a quenched population of defect-templated embryos, present in the flake before the loop begins, each of which goes supercritical when the critical radius  $R^* = \lambda/|\Delta f_A|$  falls to its own size, with no mobility factor to pay. At 143 K that radius is 0.7 to 1.4 nm, twenty to ninety cobalt sites, and the assumption carries an explicit condition that should be stated with it: the embryo distribution must be cut off above about 1.4 nm, since one 2 nm embryo anywhere in the flake would go supercritical near 200 K and sweep ten micrometers. Such a cut-off is natural for patches templated by the defect population that exfoliation leaves behind, whose sizes are set by the defects and not by the temperature, and unnatural for thermal fluctuations of the parent phase. On warming, nucleation has in either case already been paid for, the only remaining barrier is the one that controls growth, and that is exactly the situation Eq. (3) describes with a single  $E_a$ . This is cold crystallization, and it accounts for the branch asymmetry with no electronic ingredient whatever. Figure 1 draws the two paths.

That assumption is the weakest joint in the account, and it converts into two predictions, both of them cheap. If the seeds are athermal the onset cannot depend on how fast the flake was cooled, whereas thermally born seeds accumulate as  $n_0 \propto 1/|r_{\text{cool}}|$  and would raise the onset by 5.8 K per decade of cooling rate, the faster cooldown delivering the fewer seeds and hence the higher onset; one device and one afternoon settle

the provenance of the seeds. And because the area swept on the cooling branch grows as  $|r_{\text{cool}}|^{-3}$ , a cooldown slower than about 0.1 K/min must transform the flake on the way down and erase the warming anomaly altogether, which is the sharpest consequence of the picture that nobody has yet looked for.

The picture also makes an isothermal prediction. With seeds already present, a hold anywhere on the warming branch is growth limited and its half-conversion time is the radius the warming branch needs divided by the growth velocity, so a hold at 141 to 146 K, six to eleven kelvin below the onset, should complete in between three minutes and two hours, centrally between sixteen and forty-six minutes, and a hold that produces nothing there is genuinely falsifying. A hold between 150 and 170 K on the way down is a different measurement, because it probes the nucleation rate rather than growth, and the model supplies only a lower bound on its timescale, at least one to ten hours at 1 K/min; a null result there excludes nothing. The sister compound supplies the empirical anchor the model cannot: in  $\text{Na}_x\text{CoO}_2$  the alkali sublattice orders isothermally with a measured time constant near 40 min at 200 K, and a quench below 212 K suppresses the ordering entirely [36].

#### D. The high-temperature collapse

Above the formation window the ordered phase is in quasiequilibrium, since its relaxation time at 240 K is at most  $3 \times 10^{-4}$  s, shorter than any laboratory sweep by seven orders of magnitude, so the decay of the anomaly is thermodynamic rather than kinetic. Combining Eq. (2) with the equilibrium fraction gives

$$\Delta R(T) = c R_P(T) \phi_{\text{eq}}(T, B) [s(T, B) - 1], \quad (8)$$

in which two factors shrink as the temperature rises: the transport contrast  $s - 1$  of the more resistive phase measured against an increasingly conductive parent, and the equilibrium fraction itself, which collapses at  $T_{\text{ord}}(B)$  over a range broadened by the spread of local compositions. The published traces [1] decay with an apparent activation scale far below any gap in the problem, which is what Eq. (8) leads one to expect, since it convolves three temperature dependences and none of them is an activated conductivity on its own. No gap should be read from that slope, and quantifying it calls for the raw traces rather than published figures; still less should the gap the calculations of Sec. VI return be read into it, since the modest contrast the mixture bounds permit is incompatible with activation across 0.7 eV by seven orders of magnitude and more. What is interpretable without them is the ordering of the two branches: the 5 T decay is steeper and sits higher in temperature, which is what a field that raises  $T_{\text{ord}}$  and increases the contrast produces, and which a field acting on the ionic kinetics could not produce, since that would have moved the onset.

The collapse places  $T_{\text{ord}}(0) = 240 \pm 10$  K for this flake, the raw branch difference passing through 240 to 245 K and reaching zero by 260 K, with the normalized excess of Fig. 2 collapsing about ten kelvin higher for the arithmetic reason that its denominator falls with temperature as well; the parameter that enters Eqs. (7) and (8) is the boundary itself and is fixed

by the raw difference, which is the quantity that carries no baseline in it. The bulk comparison has to be made against the right scale, because bulk  $\text{Li}_x\text{CoO}_2$  carries two. The lower one, near 130 K, is kinetic: below it the magnetization depends on how fast the crystal was cooled, and it coincides with the temperature at which  $^7\text{Li}$  motional narrowing and muon-detected diffusion report the lithium ions coming to rest [20–22]. The upper one, between 155 and 185 K, is thermodynamic: it is first order and hysteretic, and the differential scanning calorimetry that places it at 175 to 185 K returns a latent heat of 82 J/mol at  $x = 0.50$  and 272 J/mol at  $x = 0.67$  [18]. Mapped onto that pair, the flake’s kinetic edge at 152 K and the bulk kinetic scale agree with nothing adjusted, and only the thermodynamic edge is displaced, 240 K against 155 to 185 K. Which flake edge is kinetic and which is thermodynamic is therefore not in question; what needs explaining is a single shifted phase boundary.

Four candidates for that shift can be ranked, and one that is often offered should be dropped. Composition comes first and is the only one already evidenced: the ordered volume fraction in bulk crystals shrinks continuously below  $x \simeq 0.45$  and the ordering field itself is bounded near  $x = 0.48$  [18, 20], so flakes at  $x = 0.37$  to 0.48 do not sit on the bulk boundary at all but drift toward the  $\sqrt{3} \times \sqrt{3}$  arrangement at  $x = 1/3$ , whose energy scale Sec. VI computes separately. Second is the cobalt-site disorder that the exfoliation route introduces, which is a mechanism and not an analogy: exfoliation removes cobalt along with lithium [31], and four atomic percent of aluminium on the cobalt site is enough to remove the hexagonal-to-monoclinic transition of the bulk entirely [29]. Third is surface stabilization of the order, for which there is independent precedent, two groups having concluded that the 155 to 185 K anomaly of chemically delithiated material is a surface-stabilized phase of significant thermal hysteresis and aging character, absent from homogeneous potentiostatic single crystals [28, 48]. Proton co-intercalation from the acid route is fourth, with evidence on both sides, proton insertion at deep delithiation having been measured by prompt gamma activation analysis [49] while hydrogen evolution during acid delithiation argues against an exchange [19], and no proton content has been measured in these flakes; strain we keep only as a formal possibility, a flake lying unclamped on  $\text{SiO}_2$  having none measured. Dimensional confinement, on the other hand, carries the wrong sign and should not be listed: finite-size scaling depresses an ordering temperature rather than raising it by 40 percent, so what can help is a specific surface enhancement, which is the third candidate, and not confinement itself.

The model separates the two edges by their rate dependence rather than by their temperature.  $T_{\text{on}}$  is fixed by  $E_a$ ,  $v$ ,  $r$  and the seed density  $n_0$ , and composition reaches it only through  $E_a$ , so a batch series across  $x = 0.37$  to 0.48 should move it by at most about  $\pm 16$  K and should widen its rise by the same amount, while  $T_{\text{ord}}$  tracks the phase boundary and may move by tens of kelvin. A batch series therefore moves the two edges by very different amounts, and a warming-rate series moves only one of them at all.

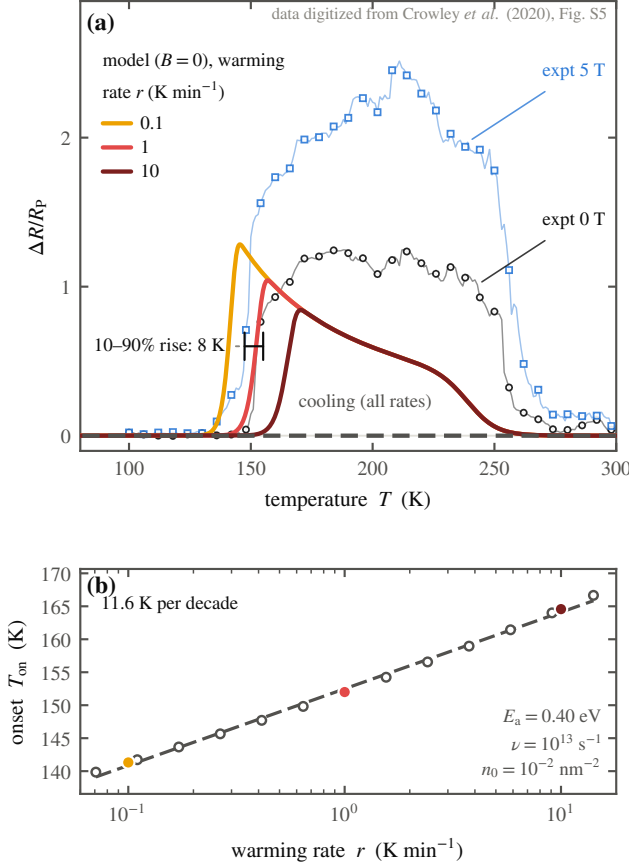


FIG. 2. Model resistance loops from the kinetic equations, set against the published traces. (a) Excess  $\Delta R/R_P$  obtained by carrying the growth velocity of Eq. (7) through the transformed fraction along a cooling and a warming sweep with  $E_a = 0.4026$  eV,  $\nu = 10^{13}$  s $^{-1}$  and  $T_{ord} = 240$  K, and converting to resistance through Eqs. (2) and (8), at three warming rates. Overlaid is the measured loop of the  $x = 0.48$  device of Ref. [1], digitized from its Fig. S5 and expressed in the same variable,  $(R_{warm} - R_{cool})/R_{cool}$ , so that the cooling branch is the zero of the plot by construction; nothing is fitted, and the model curve carries only the parameters listed above. Because that variable divides by a baseline which is itself falling with temperature, its rise midpoint at 152 K is the same as the one the raw difference  $R_{warm} - R_{cool}$  gives, while its collapse sits about ten kelvin above the 240 to 245 K through which the raw difference passes; the collapse temperatures quoted in the text belong to the raw difference and the curves plotted here to the ratio. The 5 T trace of the same figure is drawn for reference and is not described by a zero-field model. The model cooling branch, which is common to all three rates and is therefore drawn once, stays flat for as long as the sweep outruns the nose of Sec. IV C, and the bar marks the 10 to 90 percent rise width of the 1 K/min warming curve, 7.7 K about an onset of 152 K. The model peak sits somewhat above the  $c\phi(s-1) \approx 1$  of Sec. III because it is integrated at fixed contrast rather than normalized to that condition. (b) Rise midpoints against  $\log_{10} r$  over more than two decades of warming rate at fixed cooling protocol. The line is Eq. (6) with the thermodynamic factor retained, giving 11.6 K per decade at  $E_a = 0.4026$  eV and containing no parameter beyond the barrier already fixed by the onset itself; the collapse temperature, being thermodynamic, does not move at all. Separating these two slopes is the measurement that distinguishes a kinetic onset from an equilibrium one.

## V. THE MAGNETIC FIELD AND THE ORDERED PHASE

### A. What the shifted collapse implies

Treat the collapse as a first-order line in the temperature-field plane, defined by  $\Delta f(T_{ord}, B) = 0$ . Equating the differentials of the two free energies along that line, with entropy and magnetization per cobalt, gives the Clausius–Clapeyron relation in its magnetic form,

$$\frac{dT_{ord}}{dB} = \frac{\Delta M(B)}{\Delta S}, \quad \begin{aligned} \Delta M &\equiv M_O - M_P, \\ \Delta S &\equiv S_P - S_O, \end{aligned} \quad (9)$$

derived as Eqs. (S30)–(S32) of Ref. [25]. The latent entropy  $\Delta S$  is positive because the ordered phase is the one with fewer configurations available to it, so an upward shift of the collapse in field requires  $\Delta M > 0$ : the ordered, more resistive phase would then be the one carrying the larger magnetization. That is the opposite of what happens in the manganites, where a field melts charge order because the competing metallic phase is the ferromagnet [50, 51]; here the field stabilizes the resistive phase, so the moments have to live inside it.

The measured shift,  $\Delta T_{ord}/\Delta B = 1.4 \pm 1.4$  K/T, converts with  $\mu_B/k_B = 0.672$  K/T into

$$\Delta M = (2.1 \pm 2.1) \frac{\Delta S}{k_B} \mu_B \text{ per Co.} \quad (10)$$

The error bar is dominated by the uncertainty in reading the collapse from published figures, and at one standard deviation the shift is consistent with zero, so its sign is not yet established. Our own digitization of the same figure moves that conclusion a little and does not overturn it. Taking the midpoints of the two falls in the raw branch difference gives a shift of  $5 \pm 2$  K over 5 T, that is  $1.0 \pm 0.5$  K/T, which agrees with the published reading and reaches two standard deviations from zero; taking them in the excess normalized to the cooling baseline gives only  $2 \pm 2$  K, that is  $0.4 \pm 0.4$  K/T, which does not. The sign is positive in both conventions and the magnitude is smaller than the published value in both, so nothing downstream needs adjusting, since every bound in this section is derived from the larger reading and is therefore the tighter one, and the honest summary is that the digitization strengthens the sign determination in the raw difference without settling it in the ratio. What the present data do establish robustly is the other field fact, that 5 T roughly doubles the amplitude of the excess while leaving the cooling baseline within a few percent of itself, and it is the series at 1, 2, 3, 5, 7 and 9 T listed in Sec. VII that turns the chain from conditional into quantitative.

Equation (10) constrains a ratio of densities. Both  $\Delta M$  and  $\Delta S$  are per cobalt of the phase that transforms, so what the measured slope means depends on where the transformation lives, and the two possibilities differ by two orders of magnitude. Read as a bulk transformation, the ceiling is set by the  $\text{Co}^{4+}$  count: at  $x \approx 1/2$  half the cobalt sites are  $\text{Co}^{4+}$  in a low-spin  $d^5$  configuration carrying  $0.97 \mu_B$  apiece, as Sec. VI confirms, so the largest magnetization available is  $0.48 \mu_B$  per Co with every moment aligned, and Eq. (10) bounds the latent entropy at  $\Delta S \lesssim 0.23 k_B$  per Co, about a third of the



$k_B \ln 2$  that complete lithium order and disorder at half filling would release. That reading carries a difficulty of its own, which Sec. VI sets out and does not resolve: the computed bulk ground states carry no net moment, so the ceiling is never approached and a purely bulk account of the field shift is not available in the form the calculation returns.

Read instead as a transformation whose magnetization resides in the surface electron gas, the same slope means something different. A flake 20 to 60 nm thick holds forty to a hundred and twenty  $\text{CoO}_2$  layers between its two surfaces, so a magnetization of order one Bohr magneton per surface cobalt is about a hundredth of that per cobalt of the whole sample, and the latent entropy per bulk cobalt implied by Eq. (10) falls by the same factor. The bulk reading predicts a latent entropy of order a tenth of  $k_B$  per cobalt, which is a latent heat  $T\Delta S = 2.1$  meV per cobalt, 0.20 kJ per mole of cobalt, or 2.1 J/g, some twenty times a routine differential-scanning-calorimetry sensitivity of 0.1 J/g and of the same order as the 82 and 272 J/mol measured on bulk crystals at  $x = 0.50$  and  $x = 0.67$  [18]; the surface reading predicts 0.02 J/g, five times below that sensitivity. Since the two also make opposite predictions for how the excess scales with thickness, calorimetry and the thickness series form a discriminating pair rather than two separate checks, and Sec. VII states them that way.

Neither reading allows the parent metal to be the source. Pauli paramagnetism cannot supply a magnetization of this size, and the magnetic phase diagram of bulk  $\text{Li}_x\text{CoO}_2$  places the boundary between a Pauli-paramagnetic and a Curie-Weiss metal between  $x = 0.35$  and  $x = 0.40$  [18], overlapping the composition window in which the anomaly appears. The transformation is in either case a partial ordering, since the entropy it releases is at most about a third of the  $k_B \ln 2$  that complete lithium order and disorder at half filling would give. Two nearby statements should be kept apart from that one. The broadened rise and the compositional spread of the batches are a width and not an entropy, and the near-degenerate manifold of stripe variants found in Sec. VI is sub-extensive, accounting for the domain texture of the ordered phase and for no part of the entropy deficit.

### B. How many moments the field addresses

The magnetization inferred from Eq. (10) cannot be delivered by independent spins. At 5 T the Zeeman energy of a single  $g = 2$  moment is  $g\mu_B B = 0.579$  meV, while the thermal energy over the window in which the field effect operates runs from  $k_B T = 12.9$  meV at 150 K to 20.7 meV at 240 K, so a single spin is outmatched by a factor between 22 and 36 and is polarized to about two percent. What follows from that is a question of polarization rather than of energy balance, since Eq. (9) has already exhausted the free-energy comparison. A block of  $N$  moments locked parallel by exchange polarizes as  $m = \rho \tanh(N\mu_B B/k_B T)$ , with  $\rho = 0.48 \mu_B$  per cobalt the moment density that Sec. VI computes, and requiring it to reach the magnetization  $2.1\sigma$  that Eq. (10) demands of a

latent entropy  $\sigma = \Delta S/k_B$  per cobalt gives

$$N \simeq \frac{k_B T}{\mu_B B} \operatorname{arctanh}\left(\frac{2.1\sigma}{\rho}\right), \quad (11)$$

which at 5 T and 150 K, where the ordered phase forms, returns  $N \simeq 10$  for  $\sigma = 0.05$  and  $N \simeq 21$  for  $\sigma = 0.10$ , rising to 16 and 34 at the 240 K collapse. The number to carry forward is that the field-coupled object holds of order ten aligned moments rather than one, with a range of five to twenty-five. The ceiling  $\sigma \lesssim 0.23$  used above is not an independent fact about the material but the  $N \rightarrow \infty$  limit of Eq. (11), the point at which the block is fully polarized and further moments buy nothing, so the two statements are two ends of one curve rather than two findings that corroborate each other. Two structures in this material could supply such a block. One is a ferromagnetically correlated  $\text{Co}^{4+}$  domain inside the charge-ordered phase, which Eq. (11) sizes at a few lattice constants across. The other is the spin-polarized two-dimensional electron gas predicted to form on  $\text{CoO}_2$ -terminated surfaces of this material [52], on whose surface angle-resolved photoemission independently finds an electronic reconstruction of the kind a polar termination demands [53], a native reservoir of polarizable moments in a band whose exchange splitting is already built in, load-bearing here for its spin physics rather than for the hopping scale it also supplies. Only the second survives Sec. VI, which finds the bulk  $\text{Co}^{4+}$  moments held antiparallel by an exchange a hundred times the 5 T Zeeman energy.

### C. Which coupling, and how to tell

The field roughly doubles the anomaly while moving the cooling baseline by a few percent and leaving the onset where it was. Read through Eq. (8), with  $\phi$  on the rising side field-independent and  $R_P$  nearly so, the doubling requires the transport contrast of the ordered phase itself to double,

$$\frac{s(T, 5 \text{ T}) - 1}{s(T, 0) - 1} \simeq 1.9 \text{ to } 2.1, \quad (12)$$

over and above the upward shift of  $T_{\text{ord}}$  already counted. Doubling an activated conductivity ratio asks for an extra  $k_B T \ln 2$  of transport gap, which is 9 meV at 150 K and 11 meV at 183 K, a laboratory-scale energy and not an implausible one for a field acting on a correlated domain of order ten moments.

Four mechanisms can produce it, and they are distinguished by their field dependence and anisotropy rather than by their size, as Fig. 3 sets out. A field that enhances the order parameter or aligns domains through their net moment gives a Brillouin-like growth of the excess that saturates, together with a nearly isotropic response. A field that stabilizes the ordered phase through an induced moment contributes  $-\frac{1}{2}\Delta\chi B^2$  to the free-energy difference and therefore grows quadratically without saturating at laboratory fields. A field acting on a parent network near its percolation threshold [54] amplifies a small change in connectivity nonlinearly, and predicts threshold behavior, sensitivity to the sweep protocol, and large device-to-device scatter. Any mechanism tied to flux through carrier

orbits carries a  $\cos \theta$  dependence on field angle, which the spin mechanisms do not. Only perpendicular-field data exist, so the angular test has never been performed. A field series at 1, 2, 3, 5, 7, and 9 T under a fixed thermal protocol, together with one in-plane run, separates all four with no adjustable parameters, and is the single most informative measurement available on this system.

## VI. MICROSCOPIC STRUCTURE FROM FIRST PRINCIPLES

The model of Secs. III to V rests on three microscopic statements, each a well-posed density-functional question. There must be a lithium ordering transition at all, with an energy scale that puts its boundary near the observed collapse. The ordered phase must carry  $\text{Co}^{4+}$  local moments, since Eq. (10) asks it for a magnetization and Eq. (11) asks that the magnetization be collective. And it must be the more resistive of the two phases, which is the one point where Eq. (2) touches the electronic structure. The first two statements come back confirmed, the third does not, and the way in which it fails changes what the transport model is allowed to claim.

The calculations are spin-polarized, in the generalized gradient approximation [55] with projector augmented-wave datasets [56] and a Hubbard  $U$  on the cobalt  $3d$  shell [57–61], as implemented in QUANTUM ESPRESSO [62, 63], with Marzari–Vanderbilt smearing [64] and plane-wave cutoffs of 60 and 480 Ry. Three cells are treated: the row-ordered  $x = 1/2$  structure of Reimers and Dahn [14, 16] with its antiferromagnetic supercells; a  $4 \times 2$  supercell of eight formula units in which the row arrangement is compared at fixed composition against a zigzag and a clumped lithium arrangement; and the  $\sqrt{3} \times \sqrt{3}$  ordered structure at  $x = 1/3$ , the composition toward which the slowly delithiated batches move. Convergence settings, cell geometries, the magnetic seeding protocol, and the caveats that attach to each number are in Sec. S5 of Ref. [25]. Table I collects the results and Fig. 4 displays them.

The Hubbard parameter is calibrated rather than chosen. At  $U = 2.5$  eV the moment settles on the lithium-rich cobalt instead of the lithium-poor one, inverting the charge order, and the  $x = 1/3$  cell is metallic; at  $U = 5.5$  eV the  $\text{Co}^{4+}$  site is driven high spin, with a moment above  $2 \mu_B$ , and the  $x = 1/2$  cell is metallic again. Only near  $U = 4.0$  eV does the calculation reproduce the low-spin  $S = 1/2$   $\text{Co}^{4+}$  and  $S = 0$   $\text{Co}^{3+}$  configuration that the chemistry of this material requires. That one requirement fixes the parameter, and every number quoted below is quoted at that value and is meaningless without it. The Hubbard term remains a stand-in for the screened exchange that a quasiparticle self-consistent  $GW$  treatment with electron-hole interactions supplies directly for the stoichiometric parent [65, 66], so the gaps reported below label the charge-ordered magnetic state and are not offered as predictions of a gap magnitude.

TABLE I. Density-functional results, at  $U = 4.0$  eV on the cobalt  $3d$  shell unless a row states otherwise. Magnetic energies are per cobalt relative to the ground state of the same cell, lithium-arrangement energies per formula unit relative to the row arrangement, and magnetic energy differences between alignments per  $\text{Co}^{4+}$ ; no spin-Hamiltonian convention is used anywhere, so the last are plain energy differences and carry their signs. Every value in this table enters the text through a single macro defined in `dft_numbers.tex`. The three blocks come from separate campaigns and are not directly comparable: the first is relaxed with the magnetic order converged, whereas the motif block is at fixed ideal geometry with ferromagnetic alignment imposed, which is why the row structure appears there with a slightly different gap. Section S5 of Ref. [25] carries the caveats, of which the most important is that  $U$  is fixed by the requirement that the low-spin state be reproduced.

| Quantity                                                                                      | Value             |
|-----------------------------------------------------------------------------------------------|-------------------|
| <i>Row-ordered <math>x = 1/2</math>, AFM ground state</i>                                     |                   |
| $\text{Co}^{4+} / \text{Co}^{3+}$ local moment ( $\mu_B$ )                                    | 0.97 / 0.00       |
| $d$ -occupation difference ( $e$ )                                                            | 0.20              |
| Co-O bond-length contrast ( $\text{\AA}$ )                                                    | 0.001             |
| Gap, AFM / FM (eV)                                                                            | 0.70 / 0.43       |
| $E_{\text{FM}} - E_{\text{AFM}}$ (meV/Co)                                                     | +28.2             |
| $E_{\text{NM}} - E_{\text{AFM}}$ (meV/Co)                                                     | +63.7             |
| $E_{\text{AFM}} - E_{\text{FM}}$ , intra-row                                                  | -56.5             |
| $E_{\text{AFM}} - E_{\text{FM}}$ , inter-row                                                  | +41.7             |
| both in meV per $\text{Co}^{4+}$                                                              |                   |
| <i>Lithium motifs at <math>x = 1/2</math>, fixed geometry, FM imposed</i>                     |                   |
| $E_{\text{zigzag}} - E_{\text{row}}$                                                          | -8.3 / +8.2       |
| $E_{\text{clumped}} - E_{\text{row}}$                                                         | +58.1 / +92.4     |
| meV per f.u., $U = 4.0 / 0$                                                                   |                   |
| Gap, row / zigzag / clumped (eV)                                                              | 0.679/0.800/0.675 |
| $D_{\text{clumped}}/D_{\text{row}}$ at $U = 0$                                                | 1.004             |
| <i><math>\sqrt{3} \times \sqrt{3}</math> <math>x = 1/3</math>, ferrimagnetic ground state</i> |                   |
| $\text{Co}^{4+}$ local moment ( $\mu_B$ )                                                     | 1.04              |
| $d$ -occupation difference ( $e$ )                                                            | 0.26              |
| Co-O bond-length contrast ( $\text{\AA}$ )                                                    | 0.003             |
| Gap (eV)                                                                                      | 1.21              |
| $E_{\text{FM}} - E_{\text{ferri}}$ (meV/Co)                                                   | +12.4             |
| $E_{\text{NM}} - E_{\text{ferri}}$ (meV/Co)                                                   | +230              |
| the same per cell (meV)                                                                       | +37               |
| $D_{\text{NM}}(E_F)$ ( $\text{eV}^{-1}/\text{cell}$ )                                         | 7.7               |

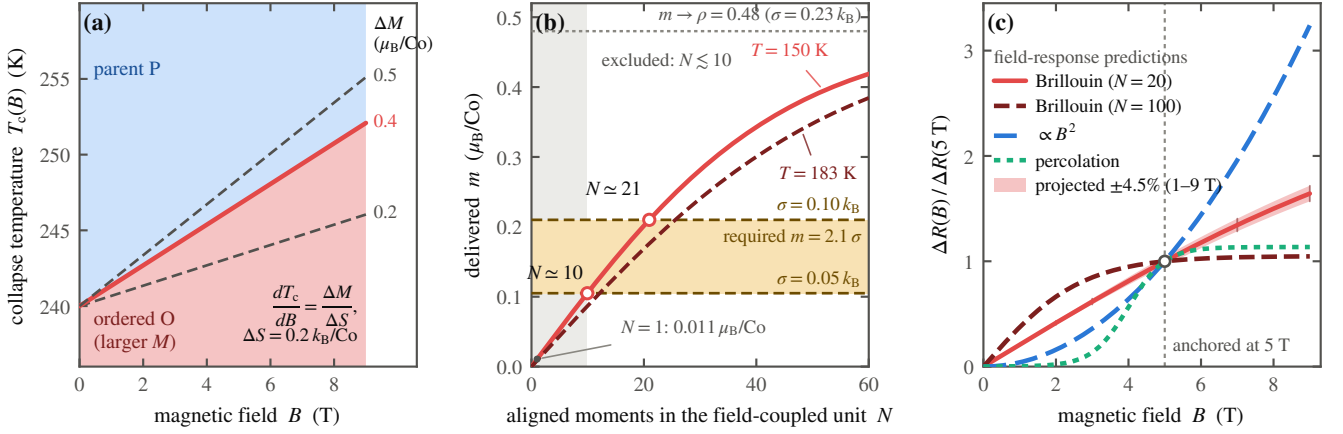


FIG. 3. Field thermodynamics of the ordered phase and the measurements that identify the coupling. (a) Clausius–Clapeyron construction of Eq. (9). The collapse line  $T_{\text{ord}}(B)$  tilts upward with slope  $\Delta M/\Delta S$ , separating the ordered phase O, which carries the larger magnetization, from the parent phase P; the contours are lines of constant  $\Delta M$  per cobalt at a latent entropy of  $0.2 k_B$  per cobalt, near the ceiling Eq. (S35) permits, and the measured shift, whose one-standard-deviation interval includes zero, selects a band of them rather than one. (b) The polarization construction of Eq. (11). The magnetization  $\rho \tanh(N\mu_B B/k_B T)$  of a block of  $N$  exchange-locked moments at 5 T and 150 K is drawn against  $N$ , with the moment density  $\rho = 0.48 \mu_B$  per cobalt as its ceiling; the horizontal lines are the magnetizations  $2.1\sigma$  that Eq. (10) requires at latent entropies of  $0.05$  and  $0.10 k_B$  per cobalt, and they are crossed at  $N \approx 10$  and  $N \approx 21$ ; the dashed curve repeats the construction at 183 K, the upper end of the anomaly, where thermal disordering pushes the same two crossings out to  $N \approx 12$  and  $N \approx 26$ . The ceiling is approached only as  $N \rightarrow \infty$ , which is the  $0.23 k_B$  bound of Eq. (S35) seen from the other end rather than an independent statement. (c) The predicted shapes of the excess amplitude against field, all normalized to agree at the one field that has been measured so that only the shape distinguishes them: Brillouin curves for aligned permanent moments at  $N = 20$  and  $N = 100$ , a quadratic curve for an induced moment, and a threshold curve for spin-dependent percolation. Saturation is weak at the smaller block size, the excess at 9 T exceeding that at 5 T by a factor 1.64 against the linear value 1.80, while  $N = 100$  gives 1.05. The shaded band is the precision the proposed series would have to reach: at  $\pm 4.5$  percent, which is  $\pm 3.2$  percent on each measured amplitude, all four cases separate at two standard deviations, whereas at a more realistic  $\pm 10$  percent the series still separates  $N = 20$  from  $N = 100$  and from the quadratic case but not from the linear limit, so resolving weak saturation is the demanding part of the measurement.

### A. Charge order, moment, and gap arrive together

In every cell examined, the charge disproportionation, the local moment, and the gap appear together or not at all. At  $x = 1/2$  the ground state is antiferromagnetic within a  $\text{Co}^{4+}$  row, carrying  $0.97 \mu_B$  on the lithium-poor cobalt against  $0.00 \mu_B$  on the lithium-rich one, a  $d$ -occupation difference of 0.20 electrons, and a gap of 0.70 eV. The moment is well determined, running from  $0.97$  to  $1.02 \mu_B$  across the ferromagnetic and both antiferromagnetic configurations, so nothing below turns on which of them is chosen. Aligning the same moments ferromagnetically costs 28.2 meV per cobalt and narrows the gap to 0.43 eV, and removing them altogether costs 63.7 meV per cobalt and closes the gap: the nonmagnetic solution of the same cell is a metal. At  $x = 1/3$  the  $\sqrt{3} \times \sqrt{3}$  structure repeats the pattern, its ground state ferrimagnetic, the two  $\text{Co}^{4+}$  moments antiparallel, with the cobalt beneath the lithium nonmagnetic and the other two carrying  $\pm 1.04 \mu_B$ , a  $d$ -occupation difference of 0.26 electrons, and a gap of 1.21 eV, while its nonmagnetic solution is a good metal with 7.7 states per eV per cell at the Fermi level lying 230 meV per cobalt higher. Forcing the moment to vanish closes the gap and erases the charge order at one stroke. The phase the transport model calls O is therefore a magnetic semiconductor and not a band insulator, and its three signatures are one transition seen from

three directions.

The moment is the one the thermodynamics asked for. Values of  $0.97$  and  $1.04 \mu_B$  are the low-spin  $S = 1/2$  result for a  $d^5$  cobalt in an octahedral field, and the companion sites sit within  $0.00 \mu_B$  of zero, which is the  $S = 0$   $d^6$  value. Half the cobalt sites at  $x = 1/2$  thus carry a full spin one-half, so the ceiling of  $0.48 \mu_B$  per cobalt that Eq. (S35) uses to bound the latent entropy of a bulk transformation is a computed quantity rather than an estimate.

The charge order is electronic and not structural. The site-mean Co-O bond lengths of the two cobalt sites differ by  $0.001 \text{ \AA}$  at  $x = 1/2$  and  $0.003 \text{ \AA}$  at  $x = 1/3$  while their  $d$  occupations differ by 0.20 and 0.26 electrons. Individual bonds are not that uniform, spanning  $0.021 \text{ \AA}$  within a single octahedron, so what is nearly absent is the contrast between the two cobalt sites rather than displacement in general; a displacive or Jahn-Teller transition carrying the same electronic contrast would separate the two sites by tens of picometers. That difference is measurable, and it becomes the spectroscopic entry of Table S2 in Ref. [25]: diffraction and phonon spectroscopy should see almost nothing across this transformation while tunneling spectroscopy sees a hard gap open.

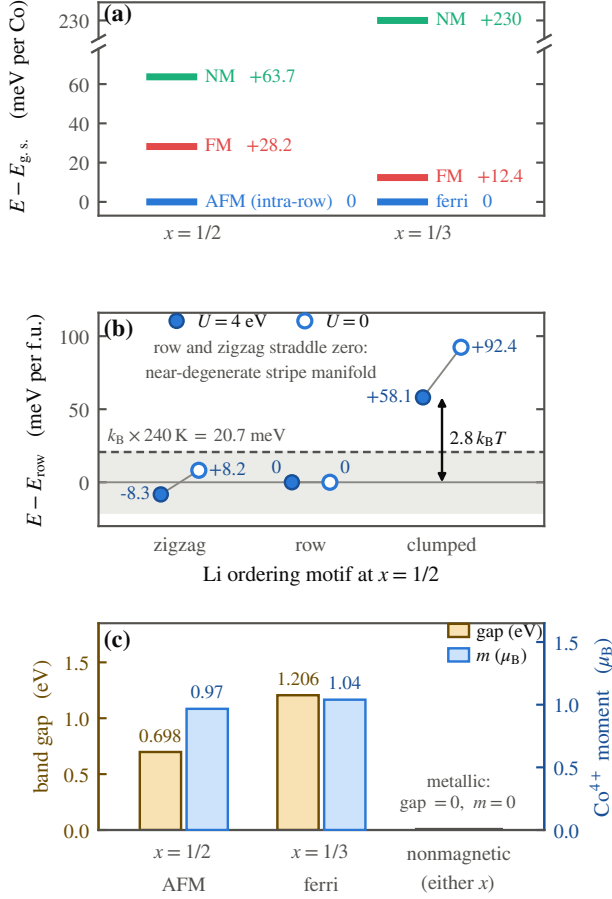


FIG. 4. Density-functional results for the ordered phases, at  $U = 4.0$  eV on the cobalt  $3d$  shell. (a) Energies of the magnetic configurations, per cobalt and relative to the ground state of each composition. At  $x = 1/2$  the antiferromagnetic alignment within a  $\text{Co}^{4+}$  row is lowest, the ferromagnetic alignment lies 28.2 meV per cobalt above it, and the nonmagnetic solution lies 63.7 meV above; at  $x = 1/3$  the ferrimagnetic arrangement is lowest, with the ferromagnetic one 12.4 meV and the nonmagnetic one 230 meV per cobalt above. (b) Energies of the three lithium motifs at  $x = 1/2$ , per formula unit and relative to the row arrangement, filled markers at  $U = 4.0$  eV and open markers at  $U = 0$ . Clumping the stripes costs 58.1 meV per formula unit, above the reference line at  $k_B \times 240$  K = 20.7 meV, while the zigzag variant changes sign between the two values of  $U$  and is therefore degenerate with the row arrangement. (c) Gap and  $\text{Co}^{4+}$  moment for each phase, 0.70 eV with  $0.97 \mu_B$  at  $x = 1/2$  and 1.21 eV with  $1.04 \mu_B$  at  $x = 1/3$ , against the nonmagnetic solutions, which have no gap at all; the panel prints the gaps to three decimals where the text rounds them to two. Moment, gap and charge order appear together, and removing the moment removes the other two.

## B. The energy scale of the ordering

Breaking the lithium stripes into clumps costs 58.1 meV per formula unit at  $U = 4.0$  eV and 92.4 meV per formula unit at  $U = 0$ , both with ferromagnetic alignment imposed and at fixed ideal geometry. That is the same order as the thermal scale of the collapse,  $k_B$  times 240 K being 20.7 meV, so a configurational excitation of a few tens of meV per formula unit places the order-disorder boundary in the few-hundred-kelvin range with nothing adjusted, and it is this energy that supplies the driving force  $\Delta f$  of Eq. (7) and the nucleation barrier of Eq. (S21). The clumped pattern is a clustering proxy and not a disordered state, so the comparison fixes an order of magnitude rather than a transition temperature.

Rerouting a stripe, by contrast, costs almost nothing at all. The zigzag arrangement lies  $-8.3$  meV per formula unit from the row arrangement at  $U = 4.0$  eV and  $+8.2$  meV per formula unit at  $U = 0$ , so the sign reverses with the Hubbard parameter and the two motifs are degenerate within the resolution of the method. The ordered phase is accordingly not one arrangement but a manifold of nearly degenerate stripe variants, which is what makes it domain textured. That degeneracy is not an entropy reservoir:  $k_B \ln 3$  per domain amounts to  $2 \times 10^{-4} k_B$  per cobalt for domains 20 nm across, four hundred times below the entropy scale the field measurement works with. What the field measurement states is the separate and stronger proposition that the transformation converts only part of the material, its latent entropy bounded by Eq. (S35) at  $0.23 k_B$  per cobalt under the bulk reading of Sec. V, which is about a third of the  $k_B \ln 2$  that complete lithium order and disorder at half filling would release.

## C. What the calculations exclude

There are two things the mechanism might have been and is not.

The first is that the resistive contrast comes from which periodic lithium motif is present, and the calculations do not support it. The comparison has to be made in the quantity that controls conduction, and since transport here is not activated across a gap, which Sec. III establishes from the size of the measured excess alone, that quantity is the density of states at the Fermi level. At  $U = 0$ , where all three arrangements are metals, their densities of states at the Fermi level agree to within one percent, the clumped-to-row ratio being 1.004 on an absolute value of 22.5 states per eV per cell, so the three motifs are indistinguishable in the one number that could have produced  $s > 1$ . At  $U = 4.0$  eV all three are instead charge-ordered insulators, with gaps of 0.679, 0.800 and 0.675 eV for the row, zigzag and clumped patterns and a density of states at the Fermi level that is numerically zero in each; that spread of gaps is not a transport statement, since a gap-activated channel would give the contrasts of  $10^7$  and larger that the measurement excludes, and what the three insulating solutions establish instead is that the charge-ordered character of the ordered phase is itself motif independent. Independent evidence points the same way, bulk  $\text{Li}_{0.5}\text{CoO}_2$  remaining metallic through its own



ordering transition, its in-plane resistivity staying between 0.75 and 0.95 m $\Omega$  cm from 4.2 to 300 K with no metal-insulator transition anywhere [28]. Whatever makes the ordered phase more resistive, it is not the lithium pattern acting on the cobalt bands, because the cobalt charge order forms equally well on top of any of these patterns and is indifferent to which. The factor  $s > 1$  in Eq. (2) cannot be assigned to a motif.

A different mechanism survives, and these calculations do not test it. All three arrangements are periodic, so a commensurate cobalt charge order is available in every one of them, whereas genuine lithium disorder is not representable in a cell this size. The natural reading is that in the kinetically frozen phase the aperiodic lithium occupation frustrates the commensurate cobalt charge order, leaving weight inside the gap and a conductivity higher than the cleanly ordered phase achieves, so that the transformation raises the resistance by removing frustration rather than by opening a gap where none existed. That reading is a hypothesis and not a result, and the price of routing the exclusion through the metallic densities of states is that the calculations then supply no estimate of  $s$  at all: the contrast of 2 to 5 is read off the measurement, and the fifth to a half of the low-energy weight that it says the ordered phase retains is the target the quasirandom calculation has to hit. The calculation that decides it is a special quasirandom structure large enough to carry an aperiodic lithium occupation at fixed  $x = 1/2$ , with its density of states at the Fermi level compared against the row-ordered cell at the same  $U$ ; the  $4 \times 2$  cell used here cannot. What the gap the calculations do return is good for is labelling the charge-ordered magnetic state, and not for setting a transport scale.

The second exclusion concerns the field, and it turns on a sign rather than on a magnitude. We report the magnetic energetics as plain energy differences per  $\text{Co}^{4+}$ , so that no spin-Hamiltonian convention is smuggled in. Within a  $\text{Co}^{4+}$  row at  $x = 1/2$  the antiferromagnetic alignment lies below the ferromagnetic one,  $E_{\text{AFM}} - E_{\text{FM}} = -56.5$  meV per  $\text{Co}^{4+}$ ; between rows the sign reverses,  $+41.7$  meV per  $\text{Co}^{4+}$ ; and at  $x = 1/3$  the ferrimagnetic ground state lies 37 meV per cell below the ferromagnetic one. The dominant coupling is antiferromagnetic at both compositions, and both computed ground states therefore carry zero net moment.

A large exchange is not in itself an obstacle to a field response; it is the very thing that would hold a cluster of twenty moments rigidly parallel so that a weak field could turn the cluster whole, and the ferromagnetically correlated  $\text{Co}^{4+}$  domain of Sec. V would be helped by it. What excludes the bulk sublattice is that its moments are not parallel: Eq. (9) asks the ordered phase for more magnetization than the parent, and an antiferromagnet offers none. The magnitude adds only the separate statement that 5 T, whose Zeeman energy is 0.58 meV against couplings near 56 meV, cannot rearrange that spin structure into one that would.

A ferromagnetically polarized reservoir is therefore required, and the most natural candidate in this material is the spin-polarized two-dimensional electron gas of  $\text{CoO}_2$ -terminated surfaces [52], polarized by construction and resident on the surfaces of a flake thin enough for surfaces to carry a large share of the conductance. It is the best of the

candidates considered here rather than the only one the material admits: oxygen  $2p$  holes, which delithiation is known to create [67], and moments on lithium-vacancy clusters, domain walls or flake edges have not been examined here, and any of them could in principle supply a polarized cluster. What distinguishes a surface reservoir from all of them is its thickness dependence, which Sec. VII turns into the measurement that decides the question.

## VII. PREDICTIONS AND DISCRIMINATING MEASUREMENTS

The model as it stands reproduces four numbers read from published traces, the onset, the collapse, the amplitude, and the field ratio, using four parameters, the barrier, the collapse temperature, the product  $c\phi(s - 1)$ , and the magnetization difference. That is a parameterization and not yet a theory; it respects every constraint, which no previous description of these data did, and it becomes a theory only through predictions that were not used in building it. Table S2 of Ref. [25] collects them, each set beside the measurement that decides it. Every entry is out of sample, and every entry can be performed on existing devices with existing instrumentation.

Three of them carry most of the weight. The warming-rate series tests the central claim that the onset is a clock reading rather than a phase boundary, and Eq. (6) predicts its slope with no freedom beyond the barrier already fixed by the onset itself; a flat onset across a factor of fifty in rate, at a cooling protocol held fixed throughout, would end the kinetic account outright. The warming-branch hold tests the growth-from-seeds picture in the most direct way available, by giving the transformation the one thing a sweep denies it, which is time at a fixed temperature. The thickness series has been promoted by the first-principles results of Sec. VI, which leave the surface electron gas as the only remaining reservoir of field-coupled moments: if the excess sheet conductance is independent of thickness, the ordered phase and its moments live on the surfaces, and if it scales with thickness, the field response needs an explanation this paper does not have.

The other entries each remove one surviving alternative: a cooling-rate series at fixed warming rate settles the provenance of the seeds, which is the one ingredient of Sec. IV C assumed rather than derived, since athermal templates leave the onset where it is while thermally born ones raise it by 5.8 K per decade; a field series at 1, 2, 3, 5, 7 and 9 T, read together with a single in-plane run, separates the three couplings of Sec. V by the shape of the amplitude alone; channel-length scaling, a four-probe measurement and a bias-history series close the contact-artifact alternative of Sec. II, including the Debye-thin space-charge layer that the blocking-contact argument leaves standing; calorimetry across the anomaly, read together with the thickness series, fixes which cobalt population Eq. (10) refers to, a bulk transformation carrying 2.1 J/g against the 0.02 J/g of a surface one; a batch series in  $x$  at fixed protocol tests whether the onset moves only through the composition dependence of  $E_a$ , so that a shift beyond about 20 K, or one uncorrelated with the width of the rise, implicates inhomogeneous

geneity or a second transition rather than a single kinetic edge; and electron or synchrotron diffraction on one flake at 100 K and again at 300 K, with weak lithium-contrast superlattice intensity expected in the warmed-through low-temperature state and absent in the as-cooled state at an interlayer spacing that does not move, confirms that what orders is the lithium sublattice and excludes the stacking glide of Sec. II A.

## VIII. DISCUSSION

Three features of this account deserve emphasis because they are what distinguish it from the description it replaces.

The first is that the onset temperature carries no information about any electronic energy scale. It is a solution of Eq. (5), a transcendental relation among a barrier, an attempt frequency, a sweep rate and a seed density, and it moves when any of the four moves, so matching such a number against an equilibrium crossover computed for an electronic model is a category error independent of how close the two numbers happen to fall. The bulk material offers the same test on better material. Its 155 to 185 K anomaly is first order, hysteretic and latent-heat bearing, and the rate dependence reported for those crystals belongs to the separate and lower lithium-freezing scale near 130 K [18, 20], so the bulk anomaly is not itself an unfreezing. What the present model predicts for it is specific: a warming-rate series on bulk crystals should move a kinetic edge near 155 K by about 12 K per decade while a thermodynamic edge stays fixed to better than  $10^{-4}$  K, which decides the character of each bulk scale on its own terms. The mapping carries a cost, in that the causal order proposed for the bulk runs opposite to the one used here, cobalt charge order below the upper scale templating the lithium order below the lower [20], whereas this account has the lithium sublattice ordering first and the cobalt charge order following. What we offer against it is the finding of Sec. VI that the cobalt charge order forms equally well on every lithium motif examined and so cannot be slaved to the lithium pattern, which reads more naturally as a cooperative transition than as one order templating the other; the same bulk warming-rate series decides that too.

The second is that the field argument has become sharper by losing a candidate, and that it remains conditional on the sign of a shift the present data do not establish. It asks only that the ordered phase carry a magnetization the parent phase does not, which Eq. (9) converts into a tilt of the collapse, and that the moments be correlated over blocks of order ten, which Eq. (11) obtains from polarization arithmetic alone. The first-principles calculations supply the moment,  $0.97 \mu_B$  of low-spin  $S = 1/2$  on half the cobalt sites, and then take away the obvious place to put it, since within the bulk those moments prefer to be antiparallel and an antiferromagnet presents nothing for a field to grip. A strong exchange would help rather than hinder, holding a block of moments rigid enough for a weak field to turn it whole, so it is the sign of the coupling and not its size that does the excluding. A ferromagnetically polarized reservoir is therefore required if the shift is real, and the spin-polarized surface electron gas [52] is the candidate named in Sec. VI, a polarized sheet of correlated moments on exactly the surfaces

that dominate a flake tens of nanometers thick.

The third is a failure that changes what the model claims, and it is reported here in full. The calculations were meant to show that the lithium-ordered arrangement has the smaller density of states at the Fermi level, and they show no such thing: at  $U = 0$ , where the comparison bears on transport at all, the row, zigzag and clumped arrangements are metals whose densities of states at the Fermi level agree to within one percent, and at  $U = 4.0$  eV all three are instead charge-ordered insulators with gaps within 0.13 eV of one another. The cobalt charge order forms on top of any lithium pattern and does not care which. What the transport model needs, therefore, is not lithium order but the difference between order and genuine disorder, and the mechanism left standing is that aperiodic lithium occupancy frustrates the commensurate cobalt charge order and leaves conducting weight in the gap, in the amount, a fifth to a half of the weight the frozen phase carries, that the measured contrast requires. That is a testable claim about a calculation, not a rhetorical retreat: a special quasirandom supercell at  $x = 1/2$  either shows in-gap weight relative to the row-ordered cell or it does not.

Scanning tunneling spectroscopy of this surface finds the ordered region conducting and the structurally disordered region gapped [35], whereas the transformation analyzed here makes the ordering raise the resistance. These are different orders and different observables, a surface cobalt registry probed by local density of states in the first case and volume lithium and charge order probed by two-terminal transport in the second. They are not in contradiction, and the frustration mechanism proposed above in fact predicts a disordered region with weight inside the gap, but only the thickness series decides which of the two the flake anomaly reports.

Beyond this material, the argument is a reminder that cathode oxides are routinely cooled through their ionic ordering temperatures at rates nobody records, and that the calorimetric tools built for glasses apply to them without modification. Nothing about the readout is peculiar to this system either, a structural transition of a mobile cation sublattice having already been followed through the two-terminal conductance of a single  $\text{Ag}_2\text{Te}$  nanowire [68]. A hysteresis loop whose warming branch differs from its cooling branch is not a curiosity to be noted and set aside; it is a measurement of a barrier, provided one records the sweep rate.

## IX. CONCLUSIONS

The warming-only resistance anomaly of slowly delithiated  $\text{Li}_x\text{CoO}_2$  nanoflakes follows from cold crystallization of the lithium sublattice. Classical nucleation and growth are never favorable together along the cooling sweep, so the flake reaches base temperature as an untransformed matrix carrying a quenched population of defect-templated embryos, which the model adopts as an assumption and which a cooling-rate series can confirm, and the warming sweep grows them into a more resistive ordered phase as soon as lithium moves. The unfreezing condition fixes an effective barrier of  $0.40 \pm 0.05$  eV, together with a seed density between  $10^{-4}$  and  $10^{-1} \text{ nm}^{-2}$ ,

which identifies the divacancy hop as the mechanism and predicts an onset shift of  $12 \pm 1$  K per decade of warming rate at fixed cooling protocol, independent of the Avrami exponent. A field-shifted collapse would require the ordered phase to carry the larger magnetization, bounding the latent entropy below about  $0.23 k_B$  per Co if the transforming phase is the bulk and by two orders of magnitude less if it is the surfaces, and the polarization arithmetic would require the field-coupled unit to hold of order ten correlated moments; the published shift is consistent with zero at one standard deviation and our digitization of the same traces returns the same sign at two standard deviations in the raw branch difference and at one in the normalized excess, so that whole chain waits on a field series.

Density-functional calculations with a Hubbard  $U$  fixed by the low-spin requirement confirm the ingredients and remove two readings. Charge order, a  $\text{Co}^{4+}$  moment of  $0.97 \mu_B$ , and a gap of  $0.70$  eV at  $x = 1/2$  and  $1.21$  eV at  $x = 1/3$  appear and vanish together, the nonmagnetic solutions being metals; clumping the lithium stripes costs  $58.1$  meV per formula unit, the same order as the thermal scale of the collapse, while rerouting them costs nothing measurable, so the transformation is a partial ordering within a nearly degenerate stripe manifold; and the charge order contrasts the site-mean Co-O bond lengths by only  $0.001$  to  $0.003$  Å, so it is electronic and nearly invisible to diffraction. Against this, the three periodic lithium motifs are indistinguishable in the density of states at the Fermi level that controls conduction, so the resistive contrast is not a matter of which motif is present, and the surviving candidate is frustration of the cobalt charge order by aperiodic

lithium, which a large quasirandom supercell can test and this work has not. The bulk  $\text{Co}^{4+}$  moments are coupled antiferromagnetically and so present no net magnetization for a field to act on, which leaves a ferromagnetically polarized reservoir as a requirement and the surface electron gas of  $\text{CoO}_2$ -terminated  $\text{Li}_x\text{CoO}_2$  as its most natural realization among the candidates considered. A warming-rate series at fixed cooling protocol, a cooling-rate series, a hold on the warming branch, a field series, and a thickness series read together with calorimetry decide the picture with measurements that need no new instrumentation.

## DATA AVAILABILITY

The scripts that generate every figure, the traces digitized from the published figures of Refs. [1, 2], and the QUANTUM ESPRESSO input files, structure files and distilled outputs behind every number in Table I are available from the author and are deposited in the repository accompanying this manuscript. The raw resistance traces underlying Refs. [1, 2] are held by the originating collaboration.

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# Supplemental Material for “Cold Crystallization of the Lithium Sublattice and the Warming-Only Resistance Anomaly of Delithiated $\text{Li}_x\text{CoO}_2$ ”

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This document gives the derivations that the main text quotes, in the order the main text uses them. Equations are numbered S1 onward and are referred to from the main text by those numbers. Section S7 lists every symbol once, with its dimension, so that no symbol carries two meanings across the two documents.

## S1. MIXTURE FORMULA AND THE TWO-PROBE CIRCUIT

### A. Bounds on a two-phase channel

Let the channel have length  $L$  and cross-sectional area  $A$ , and let a fraction  $\phi$  of its volume have transformed from the parent phase P, of resistivity  $\rho_P$ , into the ordered phase O, of resistivity  $\rho_O$ . Two microgeometries bound every other one [1, 2]. In the series arrangement the transformed material forms slabs that span the cross section, so the resistances add,

$$R_{\text{series}} = \frac{L}{A} [(1 - \phi)\rho_P + \phi\rho_O]. \quad (\text{S1})$$

With  $R_P \equiv \rho_P L/A$  the resistance the same channel would have with nothing transformed, and with the resistivity ratio

$$s \equiv \frac{\rho_O}{\rho_P}, \quad (\text{S2})$$

Eq. (S1) gives the excess exactly,

$$\frac{\Delta R}{R_P} \equiv \frac{R_{\text{series}} - R_P}{R_P} = \phi(s - 1). \quad (\text{S3})$$

In the parallel arrangement the transformed material forms filaments running the length of the channel, so the conductances add, and

$$\frac{\Delta R}{R_P} = \frac{\phi(1 - 1/s)}{1 - \phi(1 - 1/s)} \xrightarrow{\phi(1-1/s) \ll 1} \frac{\phi(s - 1)}{s}. \quad (\text{S4})$$

Both bounds have the form quoted as Eq. (2) of the main text,  $\Delta R/R_P = c\phi(s - 1)$ , with the geometry factor  $c = 1$  for the series limit and  $c = 1/s$  for the parallel limit, so that  $c \in (0, 1]$  whenever the transformed phase is the more resistive one. Both expressions vanish at  $\phi = 0$  and at  $s = 1$ , as they must, and both are monotonic in  $\phi$  and in  $s$ . The quantity the experiment supplies is the product  $c\phi(s - 1)$ , which the zero-field maximum places at approximately unity; the individual factors are not separately determined by a single trace.

### B. What the two-probe geometry allows

The measured resistance is the sum in Eq. (1) of the main text,  $R_{\text{meas}} = R_c + R_{\text{ch}}$ , where  $R_c$  collects the two Schottky junctions evaluated on the quasilinear branch of their current-voltage characteristic and  $R_{\text{ch}}$  is the channel. Writing the warming and cooling traces separately and subtracting,

$$\Delta R = [R_c^w - R_c^c] + [R_{\text{ch}}^w - R_{\text{ch}}^c], \quad (\text{S5})$$

so that any contact resistance that depends on temperature but not on thermal history cancels identically. What does not cancel is a contact whose barrier height itself depends on the lithium arrangement beneath the electrode, since that arrangement carries the same history dependence as the channel. Equation (S5) is therefore an upper bound on the channel contribution, and every quantitative statement in the main text is made about  $\Delta R$  rather than about a resistivity. The two experiments that separate the terms are channel-length scaling, under which the second bracket grows linearly with  $L$  while the first does not, and a four-probe measurement, which removes the first bracket outright.

## S2. UNFREEZING ON THE WARMING BRANCH

### A. Growth from a frozen nucleus population

Start from the growth velocity of Eq. (7) of the main text, with the sample quenched into the untransformed state and the ordered phase thermodynamically favored below  $T_{\text{ord}}$ :

$$v(T) = v_0 e^{-E_a/k_B T} q(T), \quad q(T) \equiv 1 - e^{\Delta f/k_B T}, \quad (\text{S6})$$

in which  $v_0 = \nu a$  is the velocity a domain boundary would have if every attempt succeeded,  $a = 0.28$  nm being the lithium-lithium spacing in the plane, and  $\Delta f = f_O - f_P$  is the free-energy difference per cobalt, negative below  $T_{\text{ord}}$ . The mobility factor counts the attempts that clear the migration barrier and the thermodynamic factor  $q$  removes the reverse hops, so that  $v$  vanishes at  $T_{\text{ord}}$  as it must and approaches  $v_0 e^{-E_a/k_B T}$  only at undercoolings large enough that  $|\Delta f|$  exceeds  $k_B T$ . Close to the boundary  $|\Delta f| \simeq \Delta S(T_{\text{ord}} - T)$ , so that  $q = 1 - e^{-\sigma(T_{\text{ord}} - T)/T}$  with  $\sigma = \Delta S/k_B$  per cobalt.

Nucleation is complete before the warming sweep begins, since the cooldown delivers the sample to base temperature already seeded, so the transformation is site saturated: each of the  $n_0$  nuclei per unit area grows as a disk of the same radius,

and the Kolmogorov–Johnson–Mehl–Avrami construction of Sec. S3 collapses to

$$\phi(T) = 1 - \exp[-\pi n_0 R_w(T)^2], \quad (\text{S7})$$

with  $R_w$  the radius accumulated since the start of the sweep. On a linear sweep  $T(t) = T_0 + rt$  with  $r > 0$  the substitution  $dt' = dT'/r$  turns that radius into a temperature integral,

$$\begin{aligned} R_w(T) &= \frac{1}{r} \int_{T_0}^T v(T') dT' \equiv a \Theta(T), \\ \Theta(T) &= \frac{v}{r} \int_{T_0}^T e^{-E_a/k_B T'} q(T') dT'. \end{aligned} \quad (\text{S8})$$

The integrand of  $\Theta$  has dimensions of inverse time, the integration variable has dimensions of temperature, and  $r$  has dimensions of temperature per time, so  $\Theta$  is dimensionless, as it must be if  $R_w$  is to carry the dimensions of  $a$ . Its physical meaning is the number of successful rearrangement attempts made between  $T_0$  and  $T$ , net of the reverse hops that  $q$  subtracts, and  $R_w = a\Theta$  is the distance a domain boundary advances over the same interval.

### B. Asymptotic evaluation

Change variable to  $y = E_a/k_B T'$ , so that  $T' = E_a/k_B y$  and  $dT' = -(E_a/k_B) y^{-2} dy$ . Equation (S8) becomes

$$\Theta(T) = \frac{\nu E_a}{r k_B} \int_y^{y_0} q\left(\frac{E_a}{k_B y'}\right) \frac{e^{-y'}}{y'^2} dy', \quad y = \frac{E_a}{k_B T}, \quad y_0 = \frac{E_a}{k_B T_0}. \quad (\text{S9})$$

Because  $y_0 > y$  and the integrand falls exponentially, the lower limit of the sweep contributes a fraction  $e^{-(y_0-y)}(y/y_0)^2 q(T_0)/q(T)$  of the total. For the parameters of interest,  $E_a = 0.403$  eV,  $T = 152$  K and  $T_0 = 80$  K, that fraction is  $8 \times 10^{-13}$ , so the lower limit may be replaced by infinity. The thermodynamic factor varies slowly on the scale over which the exponential does, and is carried to first order by writing the integrand as  $e^h$  with  $h = \ln q - E_a/k_B T'$ ; the pure exponential part has the standard asymptotic expansion

$$\int_y^\infty \frac{e^{-y'}}{y'^2} dy' = \frac{e^{-y}}{y^2} \left[ 1 - \frac{2}{y} + O(y^{-2}) \right], \quad (\text{S10})$$

obtained by integrating by parts once and bounding the remainder. Substituting Eq. (S10) into Eq. (S9), restoring  $y = E_a/k_B T$  and keeping the first derivative of  $q$  gives Eq. (4) of the main text together with both of its leading corrections,

$$\Theta(T) = \frac{\nu}{r} \frac{k_B T^2}{E_a} q(T) e^{-E_a/k_B T} \left[ 1 - \frac{2k_B T}{E_a} - \frac{k_B T^2}{E_a} \frac{d \ln q}{dT} + \dots \right] \quad (\text{S11})$$

The two corrections are of the same size and of opposite sign, and neither may be discarded on the ground that the other is small. At  $T = 152$  K and  $E_a = 0.403$  eV the expansion parameter is  $k_B T/E_a = 1/30.7$ , so the Laplace correction  $2k_B T/E_a$  is 6.5 percent; at  $\sigma = 0.23$  and  $T_{\text{ord}} = 240$  K the

thermodynamic factor is  $q(152 \text{ K}) = 0.125$  and  $d \ln q/dT = -0.0168 \text{ K}^{-1}$ , so the second correction is 8.3 percent with the opposite sign. The two nearly cancel in the value of  $\Theta$ , which is why the leading form suffices for the barrier, but they do not cancel in its logarithmic derivative, which is what the rate shift below measures, and they are carried separately from here on. The prefactor  $k_B T^2/E_a$  has dimensions of temperature, so  $\nu k_B T^2/r E_a$  is dimensionless.

The value of  $q$  matters as much as its derivative. Retaining it rather than setting it to unity multiplies  $\Theta$  by 0.125 at the onset, which at fixed  $n_0$  lowers the barrier the onset returns by  $k_B T_{\text{on}} \ln(1/q) = 0.027$  eV and at fixed barrier raises the nucleus density the onset demands by  $q^{-2}$ , a factor of 64. No step of the argument may treat the driving force as saturated at 152 K, since the undercooling there is only 88 K against a latent entropy of a fraction of  $k_B$ .

Two limits confirm Eq. (S11). As  $r \rightarrow 0$  the accumulated radius diverges and  $\phi \rightarrow 1$  at every temperature at which  $v$  is nonzero, which is the statement that an infinitely slow sweep is an equilibrium measurement. As  $r \rightarrow \infty$  the radius vanishes and the sample never transforms, which is the statement that an infinitely fast sweep quenches.

### C. The onset and the barrier

Define the onset as the rise midpoint,  $\phi(T_{\text{on}}) = 1/2$ , which by Eq. (S7) requires

$$\pi n_0 R_w(T_{\text{on}})^2 = \ln 2, \quad R_w(T_{\text{on}}) = \sqrt{\frac{\ln 2}{\pi n_0}}. \quad (\text{S12})$$

The onset therefore fixes not a barrier but a pair, since the same radius can be reached with a larger barrier and a denser seeding or with a smaller barrier and a sparser one; at  $n_0 = 10^{-2} \text{ nm}^{-2}$ , one nucleus per hundred square nanometres, the radius the sweep must deliver is 4.70 nm. Inserting Eq. (S11) at leading order, taking logarithms and solving for  $E_a$  gives Eq. (5) of the main text,

$$E_a = k_B T_{\text{on}} \ln \left[ \frac{\nu a k_B T_{\text{on}}^2 q(T_{\text{on}})}{E_a r R_w(T_{\text{on}})} \right]. \quad (\text{S13})$$

The argument of the logarithm is dimensionless:  $\nu$  carries  $\text{s}^{-1}$ ,  $a$  and  $R_w$  carry length,  $k_B T_{\text{on}}^2/E_a$  carries K,  $q$  is dimensionless and  $r$  carries  $\text{K s}^{-1}$ . Equation (S13) is implicit in  $E_a$  but converges under direct iteration because  $E_a$  appears inside a logarithm. The temperature to insert is the midpoint of the measured rise and not the temperature at which the excess first leaves the noise: the digitized zero-field branch difference crosses a tenth of its peak height at 149.1 K, half of it at 152.2 K and nine tenths at 163.4 K, while the same excess normalized to the cooling baseline puts the first two crossings at 149.7 and 152.7 K, so the midpoint is 152 K on either convention and the visible onset near 146 K is the foot of the rise on both;  $T_{\text{on}} = 152$  K is therefore the quantity Eq. (S12) is a statement about. Starting from  $E_a = 0.45$  eV with  $T_{\text{on}} = 152$  K,  $\nu = 10^{13} \text{ s}^{-1}$ ,  $r = 1 \text{ K/min}$  and  $n_0 = 10^{-2} \text{ nm}^{-2}$  the successive

iterates are 0.4010, 0.4025, 0.4024 eV, so three passes suffice, and numerical integration of Eq. (S8) without any expansion returns 0.4026 eV, the agreement to one part in two thousand being what remains of the two corrections in Eq. (S11) once they have cancelled against one another.

The error budget follows from differentiating Eq. (S13). Writing  $\varepsilon = E_a$ ,  $a_0 = k_B T_{\text{on}}$  and  $C = \nu a k_B T_{\text{on}}^2 q(T_{\text{on}})/(r R_w)$ , so that  $\varepsilon = a_0 \ln(C/\varepsilon)$ ,

$$\frac{\partial \varepsilon}{\partial \ln C} = \frac{a_0 \varepsilon}{\varepsilon + a_0}, \quad \frac{\partial \varepsilon}{\partial T_{\text{on}}} = \frac{\varepsilon/T_{\text{on}} + 2k_B + k_B T_{\text{on}} d \ln q/dT}{1 + a_0/\varepsilon}. \quad (\text{S14})$$

With  $\varepsilon = 0.403$  eV and  $a_0 = k_B T_{\text{on}} = 13.10$  meV, the first derivative is 12.69 meV per  $e$ -fold of  $C$ , that is 29 meV per decade; since  $C \propto \nu$  and  $C \propto 1/r$ , a decade of attempt frequency and a decade of sweep rate each move  $E_a$  by 0.029 eV, in opposite directions, and since  $C \propto R_w^{-1} \propto n_0^{1/2}$ , a decade of nucleus density moves it by half as much, 0.015 eV. The second derivative is 2.52 meV/K, whose reciprocal is the 397 K/eV that the width analysis below uses, so the  $\pm 7$  K uncertainty on the onset contributes  $\pm 0.018$  eV, and carrying the latent entropy over 0.10 to 0.69  $k_B$  per cobalt through  $q(T_{\text{on}})$  contributes  $-0.010$  to  $+0.012$  eV. Added in quadrature these give 0.049 eV, quoted as  $E_a = 0.40 \pm 0.05$  eV. Solving Eq. (S13) at the corners of the plausible ranges  $\nu \in [10^{11}, 10^{14}] \text{ s}^{-1}$ ,  $r \in [0.5, 5] \text{ K/min}$  and  $n_0 \in [10^{-4}, 10^{-1}] \text{ nm}^{-2}$  gives 0.29 eV at the low corner and 0.46 eV at the high corner, widening to the interval 0.28 to 0.48 eV once the onset and latent-entropy ranges are carried to their corners as well.

The degeneracy between the barrier and the nucleus density is exact enough to be written down. Holding  $T_{\text{on}}$  at 152 K and everything else at the values above,  $E_a = 0.403 + 0.015 \log_{10}(n_0/10^{-2} \text{ nm}^{-2})$  eV, so the plausible band  $n_0 = 10^{-4}$  to  $10^{-1} \text{ nm}^{-2}$ , corresponding to nucleus spacings of 100 down to 3 nm, maps onto  $E_a = 0.37$  to 0.42 eV. Neither number is measured on its own; what the onset measures is the combination, and the rate series of the next subsection is what separates the barrier from the seeding.

#### D. The rate shift and its independence of the Avrami exponent

Take the logarithm of the onset condition Eq. (S12), writing  $R_w = a\Theta$  and using Eq. (S11) at leading order,

$$\ln(\pi n_0 a^2) + 2 \left[ \ln \frac{\nu}{r} + \ln \frac{k_B T_{\text{on}}^2 q(T_{\text{on}})}{E_a} - \frac{E_a}{k_B T_{\text{on}}} \right] = \ln \ln 2. \quad (\text{S15})$$

Differentiate Eq. (S15) with respect to  $\ln r$  at fixed  $E_a$ ,  $\nu$  and  $n_0$ . The factor of two divides out; inside the bracket the rate term gives  $-1$ , the prefactor gives  $2 dT_{\text{on}}/(T_{\text{on}} d \ln r)$ , the thermodynamic factor gives  $(d \ln q/dT) dT_{\text{on}}/d \ln r$  and the exponential gives  $+E_a dT_{\text{on}}/(k_B T_{\text{on}}^2 d \ln r)$ , so

$$\frac{dT_{\text{on}}}{d \ln r} = \left[ \frac{2}{T_{\text{on}}} + \frac{E_a}{k_B T_{\text{on}}^2} + \frac{d \ln q}{dT} \right]^{-1}, \quad (\text{S16})$$

which is Eq. (6) of the main text and reduces, when the driving force is treated as saturated and  $d \ln q/dT$  is dropped, to the

Kissinger relation  $k_B T_{\text{on}}^2/(E_a + 2k_B T_{\text{on}})$  for a first-order rate process [3]. Numerically, at  $T_{\text{on}} = 152$  K the three terms in the bracket are 0.0132, 0.2022 and  $-0.0168 \text{ K}^{-1}$ , giving 5.04 K per  $e$ -fold, or 11.6 K per decade of warming rate; the thermodynamic factor is what carries the last of the three, and dropping it would return 10.7 K per decade. Across the barrier range that the onset admits at  $T_{\text{on}} = 152$  K the same expression gives 12.6, 11.6 and 10.8 K per decade at  $E_a = 0.37, 0.403$  and 0.43 eV, and integrating Eq. (S8) numerically over a decade of rate from 0.5 to 5 K/min gives 12.0 K. The prediction quoted in the main text is therefore  $12 \pm 1$  K per decade, which rounds outward from the 10.8 to 12.6 K span, the spread being the width of the admissible barrier range rather than a fitted uncertainty.

Suppose now that the transformed fraction is  $\phi = 1 - \exp(-A_0 \Theta^n)$  for a constant  $A_0$  and exponent  $n$ , which covers continuous nucleation with growth ( $n = 3$  in two dimensions) and growth from a frozen nucleus population ( $n = 2$ , with  $A_0 = \pi n_0 a^2$ ), as well as the simple relaxation  $n = 1$ . The onset condition becomes  $A_0 \Theta^n = \ln 2$ , whose logarithm is

$$n \left[ \ln \frac{\nu}{r} + \ln \frac{k_B T_{\text{on}}^2 q(T_{\text{on}})}{E_a} - \frac{E_a}{k_B T_{\text{on}}} \right] = \ln \frac{\ln 2}{A_0}. \quad (\text{S17})$$

Differentiating with respect to  $\ln r$  divides out the overall factor  $n$  and returns Eq. (S16) unchanged, the constant  $A_0$  dropping out entirely; direct numerical solution of Eq. (S7) at  $n = 1$  and  $n = 2$  confirms the cancellation to four figures. The onset shift per decade of warming rate is therefore a property of the barrier and the onset temperature alone, and it does not test which transformation mechanism operates. The exponent instead controls the width of the rise, which is what the next subsection quantifies, and it shifts the extracted  $E_a$  only through the constant  $A_0$  inside a logarithm, that is by the 0.014 eV per decade of  $n_0$  already budgeted.

The cancellation carries one condition, and the rate series has to be run so as to respect it. Equation (S17) treats  $A_0$ , and with it  $n_0$ , as a constant of the protocol, which it is only if the cooldown that seeds the sample is held fixed while the warming rate is varied. If the nuclei are thermally seeded then their density falls as the cooldown is hurried,  $n_0 \propto 1/|r_c|$ , and differentiating Eq. (S15) with respect to  $\ln |r_c|$  instead gives half the shift of Eq. (S16), that is  $+5.8$  K per decade of cooling rate, with the same sign, since a sparser seeding forces the sweep to grow larger domains before the midpoint is reached. A protocol that slaves the cooldown to the warming rate therefore returns about 17.4 K per decade, half as much again as the growth shift alone, and the two contributions cannot be separated after the fact. The measurement that tests Eq. (S16) is a warming series taken at a single fixed cooling rate.

#### E. Width of the rise

The bracket of Eq. (S16) is  $d \ln \Theta/dT$ , so the extended area  $A_{\text{ex}} = \pi n_0 a^2 \Theta^2$  that sits in the exponent of Eq. (S7) grows by

a factor of ten every

$$\delta T_{10\times} = \frac{\ln 10}{2} \left( \frac{d \ln \Theta}{dT} \right)^{-1} = 5.8 \text{ K} \quad (\text{S18})$$

at  $T_{\text{on}} = 152 \text{ K}$  and  $E_a = 0.403 \text{ eV}$ , half the 11.6 K that  $\Theta$  itself takes. The transformed fraction passes from 10 to 90 percent as  $A_{\text{ex}}$  runs from  $-\ln 0.9 = 0.105$  to  $\ln 10 = 2.303$ , a factor of 21.9, so

$$\Delta T_{10-90} = \delta T_{10\times} \log_{10}(21.9) = 7.8 \text{ K}, \quad (\text{S19})$$

which becomes 7.7 K, from 147.4 to 155.1 K, when Eq. (S8) is integrated across the rise rather than expanded about the midpoint, and that is the number carried forward; the corresponding width for a general exponent is larger by the factor  $2/n$ , so continuous nucleation would be narrower still and simple relaxation twice as broad. The observed width,  $19_{-9}^{+6} \text{ K}$ , exceeds all of them. A spread of barriers  $\delta E_a$  broadens the onset by  $\delta T_{\text{on}} = \delta E_a (\partial E_a / \partial T_{\text{on}})^{-1}$ , the second factor being the 397 K/eV that Eq. (S14) returns, so  $\delta E_a = 0.044 \text{ eV}$  contributes 17.4 K, which added in quadrature to Eq. (S19) gives 19.0 K. A barrier spread of that size is not an adjustable quantity here: the tetrahedral-site divacancy barrier computed for  $\text{Li}_x\text{CoO}_2$  scatters by  $\pm 0.04 \text{ eV}$  across the local lithium environments of a single composition [4], so the width of the rise and the environment dependence of the barrier are one statement rather than two. The width is accordingly consistent with growth from a frozen nucleus population carrying the environment scatter the first-principles barriers already predict, and is not consistent with a sharply cooperative transformation.

### S3. NUCLEATION, GROWTH, AND THE BRANCH ASYMMETRY

#### A. The two-dimensional nucleation barrier

A circular domain of ordered phase of radius  $R$  within the lithium plane costs line tension and gains bulk free energy [5, 6],

$$\Delta G(R) = -\pi R^2 |\Delta f_A| + 2\pi R \lambda, \quad (\text{S20})$$

with  $\Delta f_A$  the free-energy difference per unit area, negative below  $T_{\text{ord}}$ , and  $\lambda$  the domain-wall line tension, of dimension energy per length. Setting  $d\Delta G/dR = 0$  gives the critical radius and the barrier,

$$R^* = \frac{\lambda}{|\Delta f_A|}, \quad W^* = \Delta G(R^*) = \frac{\pi \lambda^2}{|\Delta f_A|}, \quad (\text{S21})$$

whose dimensions are (energy/length)<sup>2</sup> divided by (energy/area), that is energy. Close to the boundary the driving force is linear in the undercooling,  $|\Delta f_A| \approx \Delta S_A (T_{\text{ord}} - T)$  with  $\Delta S_A$  the latent entropy per unit area, so

$$\frac{W^*}{k_B T} \approx \frac{\pi \lambda^2}{k_B \Delta S_A} \frac{1}{T (T_{\text{ord}} - T)}, \quad (\text{S22})$$

which diverges as  $T \rightarrow T_{\text{ord}}^-$  and falls monotonically as the undercooling grows.

#### B. Transformed fraction

With nucleation rate per unit area  $I_n$  and growth velocity  $v$  as in Eq. (7) of the main text, the extended area fraction of the Kolmogorov–Johnson–Mehl–Avrami construction [7–10] is

$$A_{\text{ex}}(t) = \pi \int_0^t I_n(t') \left[ \int_{t'}^t v(t'') dt'' \right]^2 dt', \quad (\text{S23})$$

in which each nucleus born at  $t'$  contributes the area of the disk it would have grown by time  $t$  had it met no other domain, and the transformed fraction is

$$\phi(t) = 1 - \exp[-A_{\text{ex}}(t)]. \quad (\text{S24})$$

The dimensions check:  $I_n$  carries (area time)<sup>-1</sup>, the bracket carries length, and the outer  $dt'$  carries time, so  $A_{\text{ex}}$  is dimensionless. For constant  $I_n$  and  $v$ , Eq. (S23) reduces to  $A_{\text{ex}} = \pi I_n v^2 t^3 / 3$ , the familiar two-dimensional Avrami exponent of three. If instead nucleation is complete before growth begins, with a frozen areal density  $n_0$  of nuclei, then

$$A_{\text{ex}}(t) = \pi n_0 \left[ \int_0^t v dt' \right]^2 = \pi n_0 R_w^2, \quad (\text{S25})$$

which is Eq. (S7) and the  $n = 2$  member of the  $A_0 \Theta^n$  family analyzed in Sec. S2, with  $A_0 = \pi n_0 a^2$ ; the onset shift is then Eq. (S16) and only the width changes.

Which of the two limits applies is decided by the nucleation rate, and the point that governs the whole of this section is that the nucleation rate carries the same mobility factor as the growth velocity,

$$I_n(T) = I_0 e^{-[E_a + W^*(T)]/k_B T}, \quad (\text{S26})$$

with  $I_0 = v/a^2$  for homogeneous nucleation on the lithium sublattice and  $I_0 = Xv$  for heterogeneous nucleation on a template of areal density  $X$ . Only  $W^*$  distinguishes Eq. (S26) from Eq. (S6); the ionic mobility that has to unfreeze before a domain can grow has equally to unfreeze before a domain can be born.

#### C. Why cooling and warming differ

The two exponentials in Eqs. (S6) and (S26) move in opposite directions. By Eq. (S22) the barrier  $W^*$  is prohibitive just below  $T_{\text{ord}}$  and falls monotonically as the undercooling grows, while the mobility factor common to both rates collapses as the temperature falls. Their product has a single maximum, the nose of the time-temperature-transformation curve [11], which for every line tension considered here sits between 178 and 211 K. What it does not have is a second maximum at large undercooling: the product  $T(T_{\text{ord}} - T)$  in the denominator of Eq. (S22) is itself largest at  $T_{\text{ord}}/2 = 120 \text{ K}$ , so below that temperature  $W^*/k_B T$  rises again as the sample cools and joins the mobility factor rather than opposing it,  $dI_n/dT > 0$  holds throughout, and the bottom of the sweep is where nucleation is least likely rather than most. At 80 K the mobility factor



alone is  $e^{-56}$ , and a nucleation event there is as forbidden as a growth event.

The requirement that the smooth cooling trace places on the seeding can therefore be stated as a birth temperature. Smoothness demands that the extended area accumulated on the way down stay below some  $A_{\text{ex}}^{\text{max}}$ , which we take as  $10^{-2}$ , while the warming branch demands  $\pi n_0 R_w(T_{\text{on}})^2 = \ln 2$ ; since both are quadratic in the radius, any nucleus present at base temperature must have grown on the way down to no more than  $(A_{\text{ex}}^{\text{max}}/\ln 2)^{1/2} = 0.12$  of the radius the warming sweep will give it. The growth accumulated from a birth temperature  $T'$  downward is the same Laplace integral as Eq. (S11), evaluated at  $T'$  instead of  $T_{\text{on}}$ , so the ratio of the two radii is  $e^{-(E_a/k_B)(1/T' - 1/T_{\text{on}})}$  up to algebraic prefactors of order unity, and the condition is

$$\frac{E_a}{k_B} \left( \frac{1}{T'} - \frac{1}{T_{\text{on}}} \right) \geq \frac{1}{2} \ln \frac{\ln 2}{A_{\text{ex}}^{\text{max}}}. \quad (\text{S27})$$

The right-hand side is 2.12, and the left-hand side is steep, so the ceiling it sets is almost independent of the barrier:  $T' \leq 141.4, 142.2, 142.8$  and  $143.2$  K at  $E_a = 0.37, 0.403, 0.43$  and  $0.45$  eV. Every nucleus the sample carries to base temperature must have been born below  $T_{\text{on}} - 9$  K. A nucleus born at the nose, by contrast, grows between  $0.6$  nm and  $40$   $\mu\text{m}$  over the remainder of a  $1$  K/min cooldown against the  $1.5$  to  $47$  nm that Eq. (S12) asks of it, so a single such nucleus per flake transforms the sample on the way down and destroys the branch asymmetry.

Classical nucleation cannot deliver seeds under that ceiling, and the arithmetic says so without appeal. Pushing the maximum of Eq. (S26) below  $143$  K while  $W^*$  retains the  $1/(T_{\text{ord}} - T)$  form of Eq. (S22) requires  $|d(W^*/k_B T)/dT|$  to exceed the  $E_a/k_B T_{\text{on}}^2 = 0.20$  K $^{-1}$  that the mobility factor already supplies, hence  $W^*/k_B T \simeq 60$  at that temperature, and the rate is suppressed by  $e^{-60}$ . Put as a constraint on the one free quantity in Eq. (S22), the line tension is overdetermined: keeping the cooling trace smooth requires  $\lambda$  to be at least about  $53$  to  $59$  meV/nm over the admissible barriers, while delivering the seeding density that Eq. (S12) needs requires it to be at most  $38$  to  $48$  meV/nm. Scanning the whole of the plausible parameter space,  $E_a$  from  $0.34$  to  $0.45$  eV,  $\nu$  from  $10^{11}$  to  $10^{14}$  s $^{-1}$ ,  $\sigma$  from  $0.10$  to  $0.69$ ,  $\lambda$  over four decades and the cooling rate from  $1$  to  $60$  K/min, the two requirements never meet: the shortfall in nucleus density is five to eight decades at  $1$  K/min and never falls below  $1.9$  decades anywhere in the space, and the corner that comes closest does so only by delivering almost no nuclei at all. Heterogeneous nucleation does not rescue this. A potency factor  $\psi$  that replaces  $W^*$  by  $\psi W^*$  is exactly equivalent to replacing  $\lambda$  by  $\lambda\sqrt{\psi}$ , so it moves within the same one-parameter family already scanned, and it does so with a prefactor  $X\nu$  smaller than  $\nu/a^2$  by some three orders of magnitude, which costs a further one to one and a half decades. In the opposite limit, where the template removes the barrier altogether, the sites activate as soon as the sample crosses  $T_{\text{ord}}$ , where Eq. (S28) gives a relaxation time of  $3 \times 10^{-4}$  s, and the flake transforms on the way down.

What the argument leaves is an assumption rather than a derived result. The seeds are taken to be athermal: a quenched

population of defect-templated embryos, patches of ordered lithium pinned to structural imperfections of the flake, which are present at all temperatures and go supercritical not by a thermal fluctuation but as soon as the shrinking critical radius of Eq. (S21) falls to their own size. That crossing carries no mobility factor and therefore no  $e^{-E_a/k_B T}$  suppression, which is precisely the functional form the classical estimate cannot supply. At  $143$  K the critical radius is  $R^* = \lambda/|\Delta f_A| = 0.71, 1.07$  and  $1.43$  nm for  $\lambda = 20, 30$  and  $40$  meV/nm, that is patches of  $24, 53$  and  $94$  cobalt sites, so an embryo population whose typical member is of that size activates in the right window. The assumption carries a condition of its own: the embryo-size distribution has to be cut off above about  $1.4$  nm, because a single  $2$  nm embryo anywhere in the flake goes supercritical near  $200$  K and sweeps  $10$   $\mu\text{m}$  before the cooldown ends. A cutoff of that kind is natural for patches templated by a fixed defect population and unnatural for thermal fluctuations of the parent phase, which is the sense in which the assumption is a physical one rather than a fitted one.

The assumption is not free, because it makes two predictions that the thermal alternative does not. If the seeds are athermal then  $n_0$  is a property of the flake and not of the protocol, so  $T_{\text{on}}$  is independent of the cooling rate; if they are thermally nucleated then  $n_0 \propto 1/|r_c|$  and  $T_{\text{on}}$  moves by  $+5.8$  K per decade of cooling rate, by the differentiation of Eq. (S15) carried out in Sec. S2. One device and one afternoon separate the two. Second, the extended area accumulated on the way down scales as  $|r_c|^{-3}$  under continuous nucleation, so a cooldown slower than about  $0.1$  K/min must transform the sample on the way down and erase the warming anomaly altogether, whichever provenance the seeds have.

On the warming branch none of this is in play. The sample begins at maximum undercooling with the nucleus population already present, so Eq. (S25) applies, nucleation is not rate limiting, and the only barrier left is the one that controls growth. That is why the single-barrier description of Eq. (S6) is legitimate on warming and not on cooling, and it is why the barrier extracted from the onset is a migration barrier rather than a sum of migration and nucleation contributions.

#### D. The isothermal hold

Stopping the sweep is the sharpest test the mechanism admits, but the two branches test different things and only one of them can falsify.

A hold on the cooling branch, between  $150$  and  $170$  K, interrogates the nucleation rate, because that is the quantity the smooth cooling trace constrains. What it constrains is a lower bound. Requiring the extended area accumulated during the sweep's passage through a given temperature to stay below  $10^{-2}$ , with the Avrami exponent of three appropriate to continuous nucleation, already places  $\tau_{\text{nose}}$  above  $78$  min at  $170$  K,  $2.9$  h at  $160$  K and  $10$  h at  $150$  K for a  $1$  K/min cooldown, and above ten to a hundred minutes at  $10$  K/min. A hold of an hour that produces nothing is therefore consistent with the model as it stands; it raises the lower bound and nothing more, and a nose time of days is not excluded by any equation here.

The cooling-branch hold is worth doing, since a hold that does transform the sample fixes  $\tau_{\text{nose}}$  outright, but a null result there cannot refute the account.

The falsifying test is a hold on the warming branch, below the onset, where the nuclei are present by hypothesis and the transformation is growth limited. The half-transformation time is then the radius the onset demands divided by the growth velocity at the hold temperature,  $\tau_{1/2}(T) = R_w(T_{\text{on}})/v(T)$ , with both factors already fixed by the onset analysis. Over the admissible window this is 1 to 11 s at 176 K, 5 to 49 s at 166 K, 27 s to 4.5 min at 156 K, 3 to 34 min at 146 K and 9 min to 1.9 h at 141 K, so a hold between 141 and 146 K completes in three minutes to two hours, with the central parameters giving 46 min at the bottom of that window and 16 min at the top of it. That window is short enough to run and long enough to watch, and a hold there that produces nothing is fatal to the mechanism rather than merely uninformative. The prediction has a parameter-free anchor as well: at the onset itself the sweep has just delivered the critical radius, so  $\tau_{1/2}(T_{\text{on}}) = (k_B T_{\text{on}}^2/E_a)/r$ , which is 4 to 5 min at 1 K/min independently of  $n_0$ ,  $v$  and  $\lambda$ . Anything much longer than that at  $T_{\text{on}}$  falsifies the model without reference to any of its free parameters.

There is one empirical anchor for these timescales in the sister compound. Isothermal alkali ordering in  $\text{Na}_{0.8}\text{CoO}_2$  proceeds under a hold at 200 K with a time constant near 40 min and saturates only after two to three hours, while a quench to any temperature below about 212 K suppresses it entirely [12], which is the same phenomenology at the same order of magnitude, displaced upward in temperature as the stronger alkali-alkali interaction of the sodium system requires.

### E. Quasiequilibrium on the decay side

At the upper end of the anomaly the relaxation time of the hop that carries Eq. (S6),

$$\tau(T) = \nu^{-1} e^{E_a/k_B T}, \quad (\text{S28})$$

equals  $2.9 \times 10^{-5}$  s at  $T = 240$  K for  $E_a = 0.403$  eV and  $\nu = 10^{13} \text{ s}^{-1}$ , and  $3 \times 10^{-4}$  s at the top of the admissible barrier range, which is shorter than a full sweep at 1 K/min by more than seven orders of magnitude. The decay of the anomaly is therefore a thermodynamic collapse of  $\phi_{\text{eq}}$  together with the shrinking transport contrast, as in Eq. (8) of the main text, and not a second kinetic process.

This also fixes how far the collapse can be dragged by the sweep, which is the prediction entered in the first row of Table S2. A phase boundary crossed at rate  $r$  lags its equilibrium position by the distance the temperature moves within one relaxation time,

$$\delta T_{\text{ord}} \simeq \tau(T_{\text{ord}}) r = \nu^{-1} e^{E_a/k_B T_{\text{ord}}} r, \quad (\text{S29})$$

which at  $T_{\text{ord}} = 240$  K and the fastest rate contemplated here,  $r = 10 \text{ K/min} = 0.17 \text{ K s}^{-1}$ , is  $5 \times 10^{-6}$  K at the central barrier and  $5 \times 10^{-5}$  K at 0.45 eV. The lag is exponentially sensitive to the barrier and to nothing else, so the bound is quoted at

the top of the range rather than at the middle of it: even at 0.50 eV, above the interval the onset admits, Eq. (S29) gives  $5 \times 10^{-4}$  K. The collapse is therefore immovable below the  $10^{-3}$  K level across the whole rate series and the whole barrier range, against an onset that moves by 12 K per decade. The contrast between Eqs. (S16) and (S29) is the sharpest single statement the model makes.

## S4. FIELD THERMODYNAMICS OF THE COLLAPSE

### A. Clausius–Clapeyron relation

Let  $f_P$  and  $f_O$  be the free energies per cobalt of the two phases in a field  $B$ , with differentials  $df = -S dT - M dB$ , where  $S$  and  $M$  are the entropy and the magnetization per cobalt. The coexistence line  $T_{\text{ord}}(B)$  is defined by

$$\Delta f(T_{\text{ord}}(B), B) = 0, \quad \Delta f \equiv f_O - f_P. \quad (\text{S30})$$

Differentiating Eq. (S30) totally with respect to  $B$ ,

$$\frac{\partial \Delta f}{\partial T} \frac{dT_{\text{ord}}}{dB} + \frac{\partial \Delta f}{\partial B} = 0, \quad (\text{S31})$$

and using  $\partial \Delta f / \partial T = -(S_O - S_P) = \Delta S$  and  $\partial \Delta f / \partial B = -(M_O - M_P) = -\Delta M$ , with the definitions  $\Delta S \equiv S_P - S_O$  and  $\Delta M \equiv M_O - M_P$  of the main text, gives Eq. (9),

$$\frac{dT_{\text{ord}}}{dB} = \frac{\Delta M(B)}{\Delta S}. \quad (\text{S32})$$

The signs are worth stating explicitly. The ordered phase has fewer accessible configurations, so  $S_O < S_P$  and  $\Delta S > 0$ . The measured collapse moves upward in field, so  $dT_{\text{ord}}/dB > 0$ , and Eq. (S32) then forces  $\Delta M > 0$ : the ordered phase is the one with the larger magnetization. Dimensionally, magnetization per site divided by entropy per site is (energy/field) divided by (energy/temperature), that is temperature per field, as required.

### B. Numerical conversion

Write  $\Delta M = m \mu_B$  and  $\Delta S = \sigma k_B$  per cobalt. Then

$$\begin{aligned} \frac{dT_{\text{ord}}}{dB} &= \frac{m}{\sigma} \frac{\mu_B}{k_B}, \\ \frac{\mu_B}{k_B} &= \frac{5.788 \times 10^{-5} \text{ eV/T}}{8.617 \times 10^{-5} \text{ eV/K}} = 0.6717 \text{ K/T}. \end{aligned} \quad (\text{S33})$$

The measured secant slope is  $\Delta T_{\text{ord}}/\Delta B = (7 \pm 7) \text{ K}/(5 \text{ T}) = 1.4 \pm 1.4 \text{ K/T}$ , so

$$m = \frac{1.4 \pm 1.4}{0.6717} \sigma = (2.1 \pm 2.1) \sigma, \quad (\text{S34})$$

which is Eq. (10) of the main text.

Our own digitization of the same figure supports that slope without replacing it, and it does so with a convention attached. Reading the two collapses as the midpoints of the falls in the

raw branch difference  $R_{\text{warm}} - R_{\text{cool}}$ , which move from 243.8 to 248.9 K, gives  $5 \pm 2$  K over 5 T, that is  $1.0 \pm 0.5$  K/T; reading them in the normalized excess  $(R_{\text{warm}} - R_{\text{cool}})/R_{\text{cool}}$ , whose midpoints move from 253.4 to 255.1 K, gives  $2 \pm 2$  K, that is  $0.4 \pm 0.4$  K/T. The sign is positive in both and the magnitude is below the published 1.4 K/T in both, so Eq. (S34) and every bound that descends from it are retained at the published slope, which is the larger reading and therefore the more permissive one to bound against; what the digitization changes is only how far the sign is from zero, two standard deviations in the raw difference and one in the ratio, so the sign remains a matter for the field series rather than for a redigitization.

Everything here is per cobalt of the phase that transforms, and the bound below is quoted for the reading in which that phase is the bulk. At  $x \approx 1/2$  half the bulk cobalt sites are  $\text{Co}^{4+}$  in a low-spin  $d^5$  configuration carrying  $S = 1/2$ , so the maximum magnetization the ordered phase can carry is  $m = 0.48$  per cobalt with every such moment aligned, and the parent phase, being a Pauli-paramagnetic metal, contributes negligibly. Section S5 turns that ceiling from an estimate into a computed quantity, the ground state returning  $0.97 \mu_B$  on the  $\text{Co}^{4+}$  sites and  $0.00 \mu_B$  on the  $\text{Co}^{3+}$  sites. Equation (S34) then bounds the latent entropy,

$$\sigma \lesssim \frac{0.48}{2.1} \approx 0.23, \quad (\text{S35})$$

against the  $\ln 2 = 0.69$  per cobalt that complete lithium order and disorder at half filling would release, so under the bulk reading the transformation converts at most about a third of the available configurational entropy, which is consistent with the partial ordering that the broadened rise of Sec. S2 also indicates. Equation (S35) is not an independent statement from the headcount derived below: inverting Eq. (S37) for  $\sigma$  gives  $\sigma(N) = m_{\text{sat}} \tanh(N\mu_B B/k_B T)/2.1$ , which approaches  $m_{\text{sat}}/2.1$  as the block grows without limit, so 0.23 is the full-polarization end of the same curve whose other end fixes  $N$ . The two are an upper and a lower reading of one relation rather than two independent results. The same calculations show that this ceiling is never approached in the bulk, since the dominant coupling is antiferromagnetic,  $E_{\text{AFM}} - E_{\text{FM}} = -56.5$  meV per  $\text{Co}^{4+}$  within a row, and both computed ground states carry zero net moment, so the magnetization that tilts the collapse line has to come from elsewhere. If it resides in a surface gas, the measured slope refers to surface cobalt and the latent entropy per bulk cobalt is smaller by the surface-to-volume ratio, about  $10^{-2}$  for a flake 20 to 60 nm thick. Section VII of the main text pairs calorimetry with the thickness series to decide between the two readings.

### C. Induced moments

If the phases carry no permanent moments and respond only through their susceptibilities, the field term in the free-energy difference is  $-\frac{1}{2}\Delta\chi B^2$  rather than  $-\Delta MB$ , and Eq. (S32) integrates to

$$T_{\text{ord}}(B) - T_{\text{ord}}(0) = \frac{\Delta\chi B^2}{2\Delta S}. \quad (\text{S36})$$

The quantity extracted from a single 5 T datum is then a secant, and the terminal slope at 5 T is twice as large, so the moment demanded at 5 T becomes  $m = (4.2 \pm 4.2)\sigma$ . The induced-moment reading therefore requires more magnetization, not less, and strengthens rather than weakens the headcount of the next subsection. Equation (S36) also predicts that the collapse shift grows quadratically with field, which the field series of the main text distinguishes from the saturating shift of aligned permanent moments.

### D. Moment headcount

The question the headcount answers is one of polarization rather than of energy balance. The free-energy comparison is already exhausted by the Clausius–Clapeyron relation, which fixes the product  $\Delta M B$  against  $\Delta S dT_{\text{ord}}$  and says nothing about how many moments carry the magnetization; what the measurement additionally demands is that the ordered phase actually be magnetized by  $m = 2.1\sigma$  in units of  $\mu_B$  per cobalt at 5 T, and an independent spin cannot deliver it. At  $B = 5$  T the single-moment Zeeman energy is  $g\mu_B B = 2 \times 5.788 \times 10^{-5} \times 5 = 0.579$  meV against  $k_B T$  of 12.93, 15.77 and 20.68 meV at 150, 183 and 240 K, the onset of the ordered phase, the peak of the field-doubled excess, and the collapse, so a lone spin is outmatched by factors of 22, 27 and 36 and is polarized only to  $\tanh(\mu_B B/k_B T) = 0.022$ , 0.018 and 0.014. The field-coupled unit must therefore be a block of  $N$  moments locked parallel by exchange, whose Zeeman energy is  $N\mu_B B$  and whose polarization along the field is correspondingly larger.

Write  $m_{\text{sat}}$  for the magnetization per cobalt the ordered phase would carry with every  $\text{Co}^{4+}$  moment aligned, which Sec. S5 computes as  $m_{\text{sat}} = 0.48$ . A block of  $N$  such moments is polarized to

$$m = m_{\text{sat}} \tanh\left(\frac{N\mu_B B}{k_B T}\right), \quad (\text{S37})$$

and setting this equal to the  $m = 2.1\sigma$  that Eq. (S34) demands and solving for  $N$  gives Eq. (11) of the main text,

$$N = \frac{k_B T}{\mu_B B} \operatorname{arctanh}\left(\frac{2.1\sigma}{m_{\text{sat}}}\right), \quad (\text{S38})$$

which reduces to  $N \approx (2.1\sigma/m_{\text{sat}})(k_B T/\mu_B B)$  whenever the block is far from saturation. The criterion carries no arbitrary constant: the latent entropy  $\sigma$  enters explicitly, so that quoting a headcount requires quoting the entropy it assumes. Table S1 evaluates Eq. (S38) across the entropies the transformation admits and the three temperatures of the window.

At the latent entropy of about a tenth of  $k_B$  per cobalt that the calorimetry row of Table S2 carries, the block holds twenty-one moments at 150 K and twenty-six at 183 K; at half that entropy it holds ten and twelve. Across the entropies that a partial conversion allows, 0.02 to 0.10  $k_B$  per cobalt, and across the 150 to 183 K window in which the field-doubled excess is read,  $N$  runs from about four to about twenty-six, so the field-coupled object holds of order ten aligned moments and

TABLE S1. The size of the field-coupled block, Eq. (S38), at  $B = 5$  T and  $m_{\text{sat}} = 0.48$ , against the latent entropy  $\sigma = \Delta S/k_B$  per cobalt and the three temperatures at which the field effect is read. The last row is the ceiling  $\sigma = m_{\text{sat}}/2.1 = 0.23$  of Eq. (S35), at which the block is fully polarized and  $N$  diverges.

| $\sigma$ ( $k_B/\text{Co}$ ) | $N$ at 150 K | $N$ at 183 K | $N$ at 240 K |
|------------------------------|--------------|--------------|--------------|
| 0.02                         | 3.9          | 4.8          | 6.3          |
| 0.05                         | 9.9          | 12.1         | 15.9         |
| 0.08                         | 16.3         | 19.9         | 26.1         |
| 0.10                         | 21.0         | 25.6         | 33.5         |
| 0.15                         | 35.1         | 42.8         | 56.2         |
| 0.20                         | 60.5         | 73.8         | 96.8         |
| 0.23                         | $\infty$     | $\infty$     | $\infty$     |

the range worth quoting is five to twenty-five. Figure 3(b) of the main text draws Eq. (S37) against  $N$  at 150 K together with the magnetization the measured slope demands, and marks the crossings at  $N \simeq 10$  for  $\sigma = 0.05$  and  $N \simeq 21$  for  $\sigma = 0.10$ , the curve saturating at  $m_{\text{sat}}$ .

The energy scale required of the field-induced change in transport is a separate calculation that happens to share a logarithm. Doubling an activated conductivity ratio requires an extra

$$\delta E = k_B T \ln 2 = \begin{cases} 9.0 \text{ meV}, & T = 150 \text{ K}, \\ 10.9 \text{ meV}, & T = 183 \text{ K}, \end{cases} \quad (\text{S39})$$

of transport gap in the ordered phase, which is what Eq. (12) of the main text asks the block of Eq. (S38) to supply. The  $\ln 2$  here is the factor of two in a conductivity ratio and nothing else; it is not a criterion for  $N$ , and it is unrelated to the  $\ln 2$  that defines the rise midpoint in Eq. (S12) or to the  $\ln 2$  per cobalt of complete lithium disorder in Eq. (S35). The three appear in three different equations for three different reasons.

## S5. DENSITY-FUNCTIONAL COMPUTATIONAL DETAILS

All calculations use QUANTUM ESPRESSO 7.3.1 [13, 14] with the PBE exchange-correlation functional [15] and projector augmented-wave datasets [16] for Co, O and Li, plane-wave cutoffs of 60 Ry for the wave functions and 480 Ry for the charge density, Marzari–Vanderbilt cold smearing [17] of width 0.01 Ry, and a self-consistency threshold of  $10^{-8}$  Ry. The Hubbard term is applied to the Co 3d shell in the rotationally invariant simplified form [18] with atomic projectors.

### A. Why $U = 4.0$ eV, and why it is not a free parameter

The value is fixed by a physical requirement rather than chosen for agreement. A scan over  $U = 0, 2.5, 4.0$  and  $5.5$  eV gives four qualitatively different materials. At  $U = 0$  every cell is an itinerant metal with no site resolution. At

$U = 2.5$  eV the moment settles on the lithium-rich cobalt rather than the lithium-poor one, so that the sign of the charge disproportionation is inverted, and the  $x = 1/3$  cell is a half-metallic ferromagnet with a noninteger moment. At  $U = 5.5$  eV the  $\text{Co}^{4+}$  site is driven high spin, carrying  $2.28 \mu_B$ , and the  $x = 1/2$  cell is metallic again. Only near  $U = 4.0$  eV does the calculation return the low-spin  $S = 1/2$   $\text{Co}^{4+}$  and  $S = 0$   $\text{Co}^{3+}$  pair that the chemistry of  $\text{Li}_x\text{CoO}_2$  requires. The Hubbard parameter is therefore calibrated to a known spin state, and every number reported below carries that calibration with it. A metal-insulator transition sits between  $U = 2.5$  and  $4.0$  eV at  $x = 1/3$ ; above the transition the gap is nearly independent of  $U$ , moving from 1.179 to 1.166 eV between 4.0 and 5.5 eV while the moment continues to grow, which is the signature of a gap set by the ordering pattern rather than by  $U$  itself.

### B. Cells

The row-ordered  $x = 1/2$  structure is the one-layer monoclinic cell of the Reimers–Dahn and Van der Ven ground state [19, 20], space group  $P2_1/m$ , with two Co, four O and one Li at the experimental  $a_{\text{hex}} = 2.808 \text{ \AA}$  and  $c_{\text{hex}} = 14.24 \text{ \AA}$ . The third lattice vector carries the O3 interlayer offset, so a single  $\text{CoO}_2$  layer reproduces the ABC stacking and the cell is a one-layer setting equivalent to the reported monoclinic cell, with  $\beta_m = 108.9^\circ$ . The two cobalt sites are inequivalent by lithium coordination, two neighbors against four, which permits charge order without imposing it; they are given distinct species labels sharing one pseudopotential and one  $U$ , purely so that symmetrization cannot average them. Antiferromagnetic arrangements are treated in two supercells, one doubled along the row direction so that  $\text{Co}^{4+}$  moments alternate within a row at  $2.808 \text{ \AA}$ , and one doubled across the rows so that adjacent rows alternate at  $4.864 \text{ \AA}$ , each carrying its own ferromagnetic and nonmagnetic references so that every comparison is made inside a single cell. Sampling is  $12 \times 8 \times 8$  for self-consistency and  $16 \times 10 \times 10$  for the density of states in the primitive cell, with equivalent densities in the supercells.

The motif comparison uses a  $4 \times 2$  in-plane supercell, eight Co, sixteen O and four Li, that is twenty-eight atoms and eight formula units. A  $2 \times 2$  cell cannot do this job: on a triangular lattice at half filling all six placements of two lithium among four sites are nearest-neighbor pairs generating the same stripe order in one of three equivalent directions, so row, zigzag and clumped are literally the same structure there. In the  $4 \times 2$  cell the row and zigzag arrangements both achieve the minimum lithium-lithium coordination of two, while the clumped arrangement has four. Sampling is  $3 \times 6 \times 4$  for self-consistency and  $4 \times 8 \times 4$  for the density of states, equivalent to  $12 \times 12 \times 4$  and  $16 \times 16 \times 4$  in the primitive cell. In the row arrangement every cobalt sees exactly three lithium, so the lithium potential is uniform over the cobalt sublattice and any charge order there is spontaneous; the  $U = 0$  control confirms this, the row cell relaxing to a uniform cobalt sublattice with moments of 0.3990 and  $0.3986 \mu_B$ .

The  $x = 1/3$  calculation uses the  $\sqrt{3} \times \sqrt{3} R30^\circ$  ordering



with three Co, six O and one Li per cell, sampled at  $7 \times 7 \times 6$  and  $12 \times 12 \times 8$ .

### C. Caveats

The  $x = 1/2$  cells are built at  $c_{\text{hex}} = 14.24 \text{ \AA}$ , which rounds the  $14.2235 \text{ \AA}$  interlayer spacing reported for  $\text{Li}_{0.68}\text{CoO}_2$  [21] rather than the value at the half-lithiated composition, where the measurements give  $14.42$  to  $14.44 \text{ \AA}$  [19, 22]. The calculations therefore sit at an interlayer distance compressed by about one percent relative to the composition they describe, and since both the tetrahedral-site migration barrier and the oxygen-mediated interlayer coupling are known to respond to the  $c$  parameter, this is a geometry limitation that attaches to every number computed in the row-ordered and motif cells. No number reported here has been rescaled to compensate for it, and the comparisons that carry the argument are made between cells sharing the same geometry, so the residual enters as a common offset rather than as a contrast.

Hubbard projectors are atomic rather than orthogonalized atomic. Orthogonalized projectors were tried first and were unaffordable, a single force evaluation on the fourteen-atom cell running beyond twenty-three minutes because the derivative of the orthogonalization matrix is not accelerated in this release, and on the ten-atom cell they also prevented self-consistency from converging because two near-degenerate  $t_{2g}$  Hubbard eigenvalues rotate into each other between iterations. Every comparison reported here uses atomic projectors throughout, so the choice is internally consistent, but absolute occupation traces are projector-dependent and the charge-order claim rests on the differences between cobalt sites, which are corroborated independently by Löwdin  $d$  occupancies and by atomic-sphere charges.

The ionic relaxation of the  $x = 1/2$  cell was stopped on a force plateau. The total energy became stationary to  $0.27 \text{ meV}$  per cell over the last six steps while the reported force settled at  $78 \text{ meV/\AA}$  rather than reaching the  $25 \text{ meV/\AA}$  target. The residual sits entirely on the oxygen  $z$  coordinates in exact inversion pairs with the energy flat, so it is a systematic force-expression residual rather than a real displacement, and displacements consistent with it change the energy by less than  $0.1 \text{ meV}$ , far below the magnetic differences of tens of  $\text{meV}$  reported here. Cobalt and lithium sit on inversion centers and carry identically zero force.

The motif comparison is made at fixed ideal geometry, identical for the three arrangements, so that the comparison is internally consistent; the residual forces are nearly the same in all three and are dominated by the common Co-O breathing that accompanies charge order rather than by anything specific to the lithium configuration, so relaxation should shift the three energies together. Ferromagnetic alignment is imposed in that campaign, and antiferromagnetic alignments of the  $\text{Co}^{4+}$  moments were not explored there.

Both the motif cells and the  $x = 1/3$  cell carry one  $\text{CoO}_2$  layer. In the motif cells the third lattice vector is the true O3 rhombohedral translation, so those are bulk O3 crystals rather than slabs, but the lithium pattern is forced to repeat identically

in every layer and the specific monoclinic interlayer registry of real  $\text{Li}_{0.5}\text{CoO}_2$  is not sampled. The  $x = 1/3$  cell has the exact interlayer spacing but O1 rather than O3 registry; the gap of  $1.2 \text{ eV}$  is far larger than any plausible interlayer bandwidth and the exchange splitting is an in-plane quantity, so neither conclusion is at risk.

Runs seeded from the high-spin start prescribed at the outset, in which `starting_magnetization` =  $0.3$  places  $5.1 \mu_B$  on every cobalt because this cobalt dataset carries seventeen valence electrons, mostly failed to converge: seven of twenty-three runs were abandoned after sixty to a hundred and sixteen iterations with residuals stuck between  $10^{-2}$  and  $10^{-4} \text{ Ry}$ , and are excluded. Where such a run stabilized enough to be read, it sat about  $1.35 \text{ eV}$  per cobalt above the converged low-spin state, so the low-spin manifold is the right one by a wide margin, but the high-spin branch was not exhaustively excluded at every  $U$ .

One methodological warning is worth recording on its own. In the seven-atom primitive cell the ferromagnetic state comes out  $3.1 \text{ meV}$  per cobalt *above* the nonmagnetic one at  $U = 4.0 \text{ eV}$ , which taken alone would say the ordered phase is nonmagnetic. Both supercells say the opposite, by  $26$  to  $36 \text{ meV}$  per cobalt, and the nonmagnetic anchor itself agrees across all three cells to better than  $0.1 \text{ meV}$  per cobalt. What differs is how much symmetry the magnetic solution is permitted to break: four symmetry operations survive in the primitive cell against two in the supercells, and the supercell solution indeed carries inequivalent  $\text{Co}^{4+}$  moments of  $1.028$  and  $0.952 \mu_B$ , which the primitive cell cannot express. The same primitive cell puts the magnetic state below the nonmagnetic one at every other  $U$  tested. A direct control with symmetry switched off did not converge and is reported as inconclusive, so the conclusion rests on the agreement of the two supercells and on the cell independence of the nonmagnetic anchor. The practical lesson holds regardless: a single primitive-cell calculation would have returned the wrong sign for the magnetic instability in this material.

Finally, the disproportionation is about  $0.2 e$  rather than the  $1 e$  of a formal-valence picture, the usual situation in DFT+ $U$  charge-ordered oxides where the formal difference is screened by oxygen rehybridization. The labels  $\text{Co}^{3+}$  and  $\text{Co}^{4+}$  describe the moment cleanly and the charge only qualitatively. Gaps at  $U = 4.0 \text{ eV}$  are larger than experiment suggests for  $\text{Li}_{0.5}\text{CoO}_2$  and should not be quoted as predictions of a gap magnitude; what survives the choice of  $U$  is that the magnetic charge-ordered solutions are gapped while the nonmagnetic ones are metallic, and that the three lithium motifs are indistinguishable, the latter holding at  $U = 0$  as well.

## S6. PREDICTIONS AND DISCRIMINATING MEASUREMENTS

Table S2 collects the out-of-sample predictions that Sec. VII of the main text draws from the model, each set beside the measurement that decides it and with the numbers and the tolerances that the derivations above supply. Every entry is out of sample in the sense that no number in it was used in

fixing the four parameters of the model, and every entry can be performed on existing devices with existing instrumentation.

## S7. SYMBOLS

Table S3 collects every symbol that carries physical content in the main text and in this Supplemental Material, together with its dimension, so that a symbol appearing in a caption and the same symbol appearing inside a derivation can be checked against one another.

TABLE S2. Out-of-sample predictions of the model and what each measurement decides. Rates are warming rates; fields are applied perpendicular to the flake unless stated.

| Measurement                                                                               | Prediction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | What it decides                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Warming-rate series, 0.2 to 10 K/min, cooling protocol held fixed                         | Onset moves by $12 \pm 1$ K per decade, Eq. (6) of the main text with the thermodynamic factor retained, steepening toward 17 K per decade if the cooldown is slaved to the warm-up and the seeds are thermally born; the collapse does not move, its residual kinetic lag $\tau(T_{\text{ord}})r$ of Eq. (S29) being no larger than $5 \times 10^{-5}$ K at 10 K/min and $5 \times 10^{-6}$ K at 1 K/min anywhere in the admitted barrier range, and below $10^{-3}$ K even at 0.50 eV                                                   | A flat onset kills kinetic unfreezing; a collapse that follows the onset kills the thermodynamic assignment of $T_{\text{ord}}$ . The alkali order-disorder transition of the sister compound is sluggish enough to have been followed at more than one sweep rate [23]                          |
| Isothermal hold at 141 to 146 K on the warming branch                                     | Growth from seeds already present completes in three minutes to two hours, centrally sixteen to forty-six minutes, after which the remaining rise is absent. A hold at 150 to 170 K on the way down is a different measurement, bounded only from below at one to ten hours at 1 K/min. The measured anchor in the sister compound is a time constant of 40 min at 200 K, with ordering suppressed entirely by a quench below 212 K [12]                                                                                                  | A null result on the warming branch kills the growth-from-seeds account of Sec. IV C of the main text; a null result on the cooling branch excludes nothing                                                                                                                                      |
| Cooling-rate series, 0.05 to 10 K/min, warming rate held fixed                            | Athermal, defect-templated seeds leave the onset where it is; thermally born seeds accumulate as $1/ r_{\text{cool}} $ and raise the onset by 5.8 K per decade of cooling rate. A cooldown slower than about 0.1 K/min transforms the flake on the way down and erases the warming anomaly                                                                                                                                                                                                                                                | Settles the provenance of the seeds, which is the one ingredient of Sec. IV C of the main text assumed rather than derived                                                                                                                                                                       |
| Field series at 1, 2, 3, 5, 7, 9 T, fixed protocol                                        | Saturating Brillouin growth for aligned permanent moments, $B^2$ growth for an induced moment, threshold behavior for spin-dependent percolation                                                                                                                                                                                                                                                                                                                                                                                          | Separates the three field couplings with no adjustable parameter                                                                                                                                                                                                                                 |
| In-plane against perpendicular field                                                      | Nearly isotropic response for spin coupling, $\cos \theta$ for orbital coupling                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Selects spin against orbital coupling                                                                                                                                                                                                                                                            |
| Thickness series at fixed batch                                                           | Excess sheet conductance proportional to thickness for a volume phase, thickness-independent for a surface phase                                                                                                                                                                                                                                                                                                                                                                                                                          | Decides volume against surface, and thereby whether the surface electron gas is the ordered phase                                                                                                                                                                                                |
| Channel-length scaling, four-probe measurement, and a bias-history series                 | $\Delta R$ proportional to channel length for a channel origin and independent of it for a contact origin; amplitude independent of read voltage and of dwell time at bias if no lithium space charge accumulates at the electrodes                                                                                                                                                                                                                                                                                                       | Confirms or kills the contact-artifact alternative of Sec. II of the main text, including the Debye-thin space-charge layer that the blocking-contact argument leaves standing                                                                                                                   |
| Batch series in $x$ at fixed protocol                                                     | $T_{\text{on}}$ moves only through the composition dependence of $E_a$ , at $dT_{\text{on}}/dE_a = 397$ K/eV, so the $\pm 0.04$ eV environment scatter of the divacancy barrier [4] allows at most about $\pm 16$ K across $x = 0.37$ to $0.48$ , and the same $\delta E_a$ must appear in the width of the rise; $T_{\text{ord}}$ may move by tens of kelvin                                                                                                                                                                             | An onset shift beyond about 20 K, or one uncorrelated with the width, implicates inhomogeneity or a second transition rather than a single kinetic edge                                                                                                                                          |
| Calorimetry across the warming anomaly, read together with the thickness series           | A bulk transformation gives a latent entropy of order $0.1 k_B$ per Co, that is a latent heat of 2.1 meV per Co, 0.20 kJ/mol or 2.1 J/g, some twenty times a routine sensitivity of 0.1 J/g and comparable with the 82 and 272 J/mol measured on bulk crystals at $x = 0.50$ and $0.67$ [22], together with a relative excess independent of thickness; a surface one gives 0.02 J/g, five times below sensitivity, together with a relative excess inversely proportional to thickness                                                   | The two measurements together fix which cobalt population Eq. (10) of the main text refers to, and convert it from a ratio into a measurement of $\Delta M$ . No ordering exotherm on warming has been reported in any alkali cobaltate, so this is a new measurement rather than a confirmation |
| Diffraction or phonon spectroscopy across the anomaly, set against tunneling spectroscopy | Three separable observables. The interlayer spacing and the $(00l)$ positions do not move. The in-plane metric changes by at most a tenth of a percent, with a shear below one degree, and the lithium-contrast superlattice reflections are weak enough to require electron diffraction. The site-mean Co-O bond lengths of the two cobalt sites are computed to differ by only 0.001 to 0.003 Å, a magnitude that is strongly $U$ -sensitive and is not itself a prediction of the method, while a hard gap opens in the local spectrum | Separates electronic charge order from a displacive or Jahn-Teller transition, which would contrast the two sites by tens of picometers, and from a stacking glide, which would collapse the interlayer spacing by six percent                                                                   |
| Electron or synchrotron diffraction on one flake at 100 K and at 300 K                    | Weak lithium-contrast superlattice intensity present in the warmed-through low-temperature state and absent in the as-cooled state, with the interlayer spacing and the $(00l)$ positions unchanged to better than half a percent throughout                                                                                                                                                                                                                                                                                              | Confirms that what orders is the lithium sublattice, and excludes the O3 to H1-3 stacking glide of Sec. II A of the main text, which collapses the interlayer spacing by about six percent                                                                                                       |

TABLE S3. The symbols that carry physical content in the main text and in this Supplemental Material, with their dimensions. The left panel collects the transport and rate quantities, the right panel the nucleation, thermodynamic and electronic-structure quantities. Purely local auxiliaries introduced and discharged inside a single derivation, such as the integration variable  $y$  of Eq. (S9) and the substitutions  $\varepsilon$ ,  $a_0$  and  $C$  of Eq. (S14), are defined where they appear and are not repeated here. No symbol in the list below carries a second meaning anywhere in either document.

| Symbol                  | Meaning                                       | Dimension         | Symbol           | Meaning                                               | Dimension                       |
|-------------------------|-----------------------------------------------|-------------------|------------------|-------------------------------------------------------|---------------------------------|
| $\phi$                  | transformed volume fraction                   | dimensionless     | $I_n, I_0$       | nucleation rate per area and its prefactor            | $\text{m}^{-2}\text{s}^{-1}$    |
| $\phi_{\text{eq}}$      | equilibrium transformed fraction              | dimensionless     | $v, v_0$         | domain growth velocity and its prefactor              | $\text{m s}^{-1}$               |
| $x$                     | lithium content of $\text{Li}_x\text{CoO}_2$  | dimensionless     | $n_0$            | frozen nucleus areal density                          | $\text{m}^{-2}$                 |
| $q$                     | thermodynamic factor $1 - e^{\Delta f/k_B T}$ | dimensionless     | $a$              | lithium-lithium spacing in the plane                  | $\text{m}$                      |
| $s$                     | resistivity ratio $\rho_0/\rho_p$             | dimensionless     | $\lambda$        | domain-wall line tension                              | $\text{eV m}^{-1}$              |
| $c$                     | mixture geometry factor                       | dimensionless     | $W^*, R^*$       | nucleation barrier and critical radius                | $\text{eV, m}$                  |
| $R_{\text{meas}}$       | measured two-probe resistance                 | $\Omega$          | $A_{\text{ex}}$  | extended area fraction                                | dimensionless                   |
| $R_p$                   | channel resistance of the parent phase        | $\Omega$          | $A_0, n$         | Avrami prefactor and exponent                         | dimensionless                   |
| $R_c, R_{\text{ch}}$    | contact and channel resistance                | $\Omega$          | $\Delta f$       | free-energy difference per cobalt                     | $\text{eV}$                     |
| $\Delta R$              | warming-minus-cooling excess                  | $\Omega$          | $\Delta f_A$     | free-energy difference per unit area                  | $\text{eV m}^{-2}$              |
| $L, A$                  | channel length and cross section              | $\text{m, m}^2$   | $\Delta S$       | latent entropy per cobalt                             | $\text{eV K}^{-1}$              |
| $k(T)$                  | rearrangement rate                            | $\text{s}^{-1}$   | $\Delta S_A$     | latent entropy per unit area                          | $\text{eV K}^{-1}\text{m}^{-2}$ |
| $\nu$                   | attempt frequency                             | $\text{s}^{-1}$   | $\sigma$         | $\Delta S$ in units of $k_B$                          | dimensionless                   |
| $E_a$                   | effective migration barrier                   | $\text{eV}$       | $\Delta M$       | magnetization difference per cobalt                   | $\text{eV T}^{-1}$              |
| $\delta E_a$            | spread of barriers across the sample          | $\text{eV}$       | $m$              | $\Delta M$ in units of $\mu_B$                        | dimensionless                   |
| $r$                     | sweep rate                                    | $\text{K s}^{-1}$ | $\Delta\chi$     | susceptibility difference per cobalt                  | $\text{eV T}^{-2}$              |
| $T_0$                   | temperature at which the sweep starts         | $\text{K}$        | $B$              | applied magnetic field                                | $\text{T}$                      |
| $\Theta$                | accumulated attempts on the sweep             | dimensionless     | $g$              | spectroscopic splitting factor                        | dimensionless                   |
| $\delta T_{10\times}$   | temperature per decade of $A_{\text{ex}}$     | $\text{K}$        | $N$              | moments in the field-coupled block                    | dimensionless                   |
| $\Delta T_{10-90}$      | 10 to 90 percent rise width                   | $\text{K}$        | $\delta E$       | field-induced change of transport gap                 | $\text{eV}$                     |
| $R_w$                   | growth radius accumulated on warming          | $\text{m}$        | $m_{\text{sat}}$ | $\Delta M$ with every $\text{Co}^{4+}$ moment aligned | dimensionless                   |
| $T_{\text{on}}$         | rise midpoint of the anomaly                  | $\text{K}$        | $U$              | Hubbard parameter on Co $3d$                          | $\text{eV}$                     |
| $T_{\text{ord}}$        | first-order collapse temperature              | $\text{K}$        | $D(E_F)$         | density of states at the Fermi level                  | $\text{eV}^{-1}$                |
| $\delta T_{\text{ord}}$ | kinetic lag of the collapse                   | $\text{K}$        | $n_d$            | cobalt $3d$ occupation                                | dimensionless                   |
| $\tau$                  | relaxation time $\nu^{-1} e^{E_a/k_B T}$      | $\text{s}$        | $\beta_m$        | monoclinic cell angle                                 | degrees                         |
| $\tau_{\text{nose}}$    | transformation time at the nose               | $\text{s}$        | $\mathcal{H}$    | thermal history of the sample                         | dimensionless                   |
| $\tau_{1/2}$            | isothermal half-transformation time           | $\text{s}$        | $X$              | areal density of nucleation templates                 | $\text{m}^{-2}$                 |
| $\psi$                  | heterogeneous nucleation potency              | dimensionless     |                  |                                                       |                                 |



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